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**GazaCare Connect: Remote Medical Bridge for
External Specialist Consultations**

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وإلى الجرحى، الذين يحملون في أجسادهم شاهدًا على ما جرى، وفي عيونهم إصرارًا على أن يُكملوا الطريق،

وإلى كل من فقد عزيزًا — أبًا، أمًا، أخًا، ابنًا، أو صديقًا — وحمل الفقد في صمت، واستمر رغم الغياب الذي لا يُشفى.

لكم وحدكم، لا لأحد سوى لكم، نُهدي هذا الجهد المتواضع،

عسى أن يكون خطوة صغيرة على طريق طويل نحو غزة التي تستحق أن تُشفى، لا أن تُداوى فقط.

Abstract

The healthcare system in the Gaza Strip faces ongoing challenges in providing timely and organized access to specialized medical consultations. A persistent shortage of specialized medical professionals, combined with increasing pressure on healthcare facilities, has limited patients' ability to obtain expert medical opinions, continuous follow-up, and second medical consultations. As a result, many patients rely on fragmented or informal communication channels that lack proper organization, security, and medical accountability.

This research presents **GazaCare Connect**, an integrated telemedicine-based system designed to provide secure and structured remote medical consultations for patients residing in the Gaza Strip. The proposed system enables patients to communicate with specialized physicians located outside Gaza through a web-based platform that supports written medical consultations, secure uploading of medical reports and test results, and timely professional medical responses.

The study focuses on the design and development of the system using modern web technologies to ensure usability, reliability, and data security. GazaCare Connect incorporates user authentication, structured consultation workflows, and data protection mechanisms to safeguard sensitive medical information. The system is evaluated based on usability, effectiveness of communication, data security, and its contribution to improving access to specialized medical consultation services.

The findings indicate that GazaCare Connect has the potential to enhance access to specialized medical consultations, reduce reliance on informal communication methods, and improve the efficiency and quality of medical decision-making. By leveraging telemedicine technologies, the system offers a practical and scalable solution for supporting healthcare delivery in Gaza and highlights the role of digital health platforms in addressing healthcare challenges in constrained environments.

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CHAPTER 1 INTRODUCTION

1.1 Background and Context

The healthcare sector plays a fundamental role in ensuring the well-being of individuals and the stability of societies. Access to timely and appropriate medical consultations is a key component of effective healthcare delivery, particularly for patients who require specialized opinions, follow-up care, or second medical consultations. In certain geographical areas, such as the Gaza Strip, providing consistent access to specialized medical consultations remains a significant challenge due to long-standing political, economic, and infrastructural constraints [1].

The healthcare system in the Gaza Strip faces persistent limitations, most notably a shortage of specialized medical professionals in several medical fields, in addition to the increasing pressure on healthcare facilities caused by high population density. These factors have reduced the capacity of hospitals and clinics to meet the growing demand for specialized medical consultations and continuous follow-up services.

As a result, many patients experience delays in obtaining expert medical advice, particularly in cases that require specialist evaluation, long-term disease management, or second medical opinions. The absence of structured and reliable mechanisms for remote medical consultation further complicates access to specialized care, leaving patients dependent on limited local services or informal communication channels.

In response to these challenges, telemedicine has emerged as a practical and effective approach to improving access to medical consultations by leveraging modern information and communication technologies. Telemedicine enables patients to communicate with specialized physicians remotely, securely exchange medical information, and receive professional medical guidance without geographical constraints.

Accordingly, the GazaCare Connect project was developed as an integrated electronic platform aimed at facilitating organized and secure remote medical consultations. The system connects patients inside the Gaza Strip with specialized physicians outside the region, allowing patients to submit consultation requests, upload medical reports and test results, and receive professional medical recommendations in a timely manner. By focusing exclusively on medical consultations rather than emergency or critical

interventions, GazaCare Connect seeks to enhance the quality, accessibility, and efficiency of healthcare services in Gaza.

1.2 Research Problem

Many patients who require specialist opinions or long-term medical follow-up are forced to depend on informal communication channels, such as social media or personal contacts, which lack medical oversight, data security, and continuity of care. These practices increase the risk of misinformation, misdiagnosis, and violations of patient privacy.

Therefore, the core research problem addressed in this study is the absence of a secure, organized, and integrated electronic platform that enables patients in the Gaza Strip to access remote specialized medical consultations in a reliable and confidential manner.

1.3 Research Questions

- How can a secure electronic platform enable patients in the Gaza Strip to access remote specialized medical consultations?
- To what extent can a telemedicine system help address the shortage of specialized medical consultation services?
- How can remote medical consultations improve the timeliness of obtaining medical advice?
- How can the system enhance the accuracy of medical decision-making through organized specialist consultations?
- How effective is the proposed system in organizing communication between patients and physicians?
- How can GazaCare Connect support follow-up and continuity of care for patients requiring ongoing consultations?
- What functional requirements are necessary to support secure and reliable remote medical consultations?
- What security and privacy mechanisms are required to protect patients' medical data?
- What is the level of usability and acceptance of the GazaCare Connect system among patients and physicians?

- How can the proposed system contribute to improving the overall efficiency and quality of healthcare services in the Gaza Strip?

1.4 Research Objectives

The main objective of this research is to design and develop an integrated telemedicine system that improves access to specialized medical consultations for patients in the Gaza Strip.

- To design a secure electronic platform that enables remote specialized medical consultations.
- To improve patient access to specialized medical opinions regardless of geographical constraints.
- To support timely delivery of medical advice through organized digital consultations.
- To enhance the accuracy of medical decision-making by facilitating structured specialist evaluations.
- To organize and streamline communication between patients and physicians.
- To support continuity of care through follow-up consultations and medical record management.
- To define and implement essential functional requirements for remote medical consultation systems.
- To ensure high levels of data security, privacy, and confidentiality for medical information.
- To evaluate the usability and acceptance of the system among patients and physicians.
- To contribute to improving the efficiency and overall quality of healthcare services in the Gaza Strip.

1.5 Research Significance

1.5.1 Human Significance

The system reduces patient suffering by enabling timely access to specialized medical advice, improving health outcomes and quality of life.

1.5.2 Scientific Significance

The research contributes to academic knowledge by presenting a practical telemedicine model for constrained and crisis-affected environments.

1.5.3 Technical Significance

It provides a reference model for secure, scalable, and user-friendly telemedicine platforms.

1.5.4 Societal Significance

The system reduces hospital overcrowding and promotes efficient use of healthcare resources.

1.5.5 Policy and Strategic Significance

The research offers valuable insights for policymakers and NGOs on integrating telemedicine into healthcare strategies.

1.6 Scope and Limitations

1.6.1 Scope

- **Geographical:** Patients in the Gaza Strip and physicians outside Gaza.
- **Functional:** User registration, consultation submission, medical file uploads, physician responses, and data management.
- **Technological:** Web-based telemedicine platform

1.6.2 Limitations

- Internet and infrastructure constraints
- Legal and data privacy regulations
- No replacement for emergency or in-person care
- Context-specific findings

1.7 Summary

Chapter One has provided a comprehensive introduction to the GazaCare Connect project, focusing on the critical challenges faced by the healthcare system in the Gaza Strip. The background and context highlighted the severe shortage of specialized medical professionals, limited technical resources, and the difficulties patients face in accessing timely medical care.

The research problem was clearly identified as the absence of a reliable and integrated platform to provide secure and organized remote medical consultations, bridging the gap between patients in Gaza and specialized physicians outside the region. Based on this problem, the research formulated a set of research questions that guide the study and ensure that all aspects of healthcare access, quality, and system functionality are addressed.

CHAPTER 2 LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature on telemedicine systems, highlighting their effectiveness in improving healthcare access, especially in resource-limited and crisis-affected regions. It examines technologies, security mechanisms, and previous projects relevant to GazaCare Connect.

2.2 Techniques

This section describes the main tools, technologies, and techniques used in the design and development of the *GazaCare Connect* system. The selection of these technologies was based on their reliability, security, scalability, and suitability for building modern web-based healthcare applications in resource-constrained environments.

2.2.1 Design Techniques

Figma: Figma is a cloud-based User Interface (UI) and User Experience (UX) design tool used to design the visual layout of the GazaCare Connect system. It enables designers and developers to collaborate in real time while creating wireframes, mockups, and interactive prototypes for web applications.

Figma supports cross-platform usage through web and desktop applications on Windows and macOS. It also provides reusable components and ensures design consistency across system pages. In addition, Figma facilitates the handoff process between designers and developers by providing detailed design specifications that help translate visual designs into functional code [2].



Figure 1 Figma

Edraw Max: Edraw Max is a professional two-dimensional diagramming software used to create flowcharts, organizational charts, workflow diagrams, and system architecture diagrams. In this project, Edraw Max is utilized to model system workflows, administrative processes, and the overall system structure.

The tool supports clear visualization of complex processes, which assists in system analysis, improves understanding of functional requirements, and enhances the quality of project documentation [3].



Figure 2 Edraw Max

Draw.io: Draw.io is a free online diagramming tool that supports the creation of flowcharts, UML diagrams, Entity Relationship (ER) diagrams, and data flow diagrams. It provides a simple drag-and-drop interface that allows rapid development of detailed technical diagrams.

In the GazaCare Connect project, Draw.io is used to design database schemas, data flow models, and interaction diagrams between system components. The tool also supports cloud integration, making it suitable for collaborative documentation and system modeling [4].



Figure 3 Draw.io

2.2.2 Web Techniques

2.2.2.1 Front-End Techniques

The front-end of the GazaCare Connect system is responsible for delivering an interactive and user-friendly interface for patients, physicians, and administrators. It focuses on usability, responsiveness, and accessibility across different devices. The following front-end technologies were used:

React.js: React.js is a JavaScript-based library used for building dynamic and interactive user interfaces. It enables the development of reusable UI components and efficient rendering of content based on user interactions. React.js improves performance and maintainability of the user interface through its component-based architecture [5].

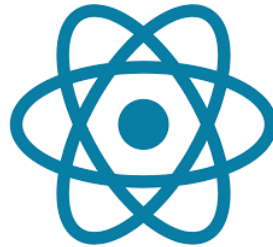


Figure 4 React.js

Next.js: Next.js is a React-based framework used to enhance performance, scalability, and search engine optimization (SEO). It supports server-side rendering (SSR) and static site generation (SSG), which improve page loading speed and overall system performance. In GazaCare Connect, Next.js is used to structure the front-end application and improve user experience [6].



Figure 5 Next.js

HTML5: Hypertext Markup Language (HTML5) is the standard markup language used to structure web pages. It defines the content and layout of the system's web interface and supports multimedia elements. HTML5 ensures compatibility across modern web browsers and forms the foundation of the front-end structure.



Figure 6 HTML5

CSS3: Cascading Style Sheets (CSS3) is used to control the presentation and layout of web pages. It enables the separation of content and design, allowing consistent styling, responsive layouts, and improved accessibility. CSS3 enhances the visual appearance of the GazaCare Connect platform through flexible design properties such as colors, fonts, and layouts [7].



Figure 7 CSS3

TypeScript: TypeScript is a strongly typed superset of JavaScript that adds static typing and enhanced error checking during development. It improves code reliability, readability, and maintainability, especially in large-scale applications. In this project, TypeScript is used to reduce runtime errors and enhance the robustness of the front-end codebase [8].



Figure 8 TypeScript

Visual Studio Code: Visual Studio Code is an integrated development environment (IDE) used for writing, editing, and debugging source code. It supports multiple programming languages, including TypeScript, HTML, and CSS, and provides extensions that enhance productivity, code formatting, and error detection [9].

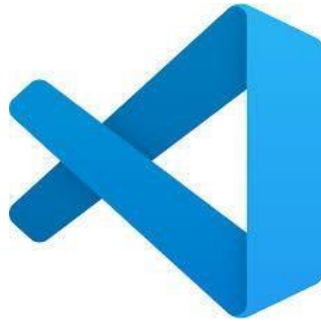


Figure 9 VS Code

2.2.2.2 Back-End Techniques

The back-end of the GazaCare Connect system is responsible for managing system logic, handling user requests, processing data, and ensuring secure communication between the front-end and the server. It also manages user authentication, stores medical consultations, and securely handles medical reports.

The following back-end technologies were used:

Laravel: Laravel is a PHP-based web application framework used to build secure and scalable back-end systems. It provides built-in features such as routing, authentication, middleware, and database management. Laravel is suitable for managing complex systems like telemedicine platforms due to its structured architecture and security capabilities [10].

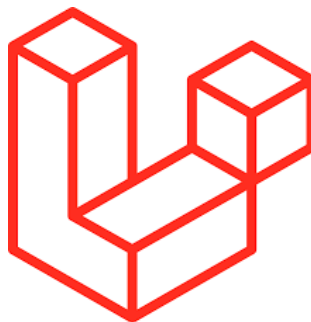


Figure 10 Laravel

PHP: PHP is the server-side programming language used by Laravel to handle application logic, process requests, and interact with the database. It plays a central role in managing system operations and ensuring smooth data flow between system components [11].



Figure 11 PHP

MySQL Database: MySQL is a relational database management system used to store patient data, medical consultations, and test results securely. It ensures data integrity, structured storage, and efficient querying, making it suitable for healthcare applications that require reliable data management [12].



Figure 12 MySQL Database

phpMyAdmin: phpMyAdmin is a web-based database management tool used to manage MySQL databases. It allows administrators to visualize database tables, execute queries, and manage relationships between data entities easily and efficiently [13].



Figure 13 phpMyAdmin

2.3 Previous Studies

This section reviews several projects and platforms related to telemedicine and remote healthcare services that are similar to the proposed *GazaCare Connect* system. The

review provides an analysis of their main features, advantages, and limitations in order to highlight the competitive value and unique contributions of the proposed system. By examining existing solutions, this section identifies the gaps that GazaCare Connect aims to address and emphasizes the need for a comprehensive and secure telemedicine platform tailored to the healthcare challenges faced by patients in the Gaza Strip.

2.3.1 Gxza Health Overview

Gxza Health is a web-based healthcare platform designed to serve patients in the Gaza Strip. The platform primarily focuses on providing general health information and limited remote medical consultation services [14].

Advantages

- Provides online health education and reliable medical resources.
- Features a simple and user-friendly interface.
- Accessible to patients in Gaza with internet connectivity.
- Helps reduce unnecessary hospital visits for minor or routine medical cases.
- Offers relatively quick responses to general health inquiries.
- Encourages patient engagement and awareness of health-related issues.
- Accessible via both desktop and mobile devices.
- Enables communication between patients and doctors without requiring physical presence.

Disadvantages

- Limited availability of specialized medical professionals.
- Lacks a structured mechanism for uploading medical reports or maintaining patient medical history.
- Does not support advanced features such as case severity classification or centralized consultation management.
- The platform may temporarily stop accepting new consultations during periods of high demand, limiting patients' access to healthcare services.
- The consultation process involves third parties, which may reduce patient privacy compared to direct communication between patients and doctors.
- Limited Responsive Design for Mobile.



Figure 14 Gxza Health Overview

2.3.2 MENA Health Overview

MENA Health is a telehealth platform that provides healthcare services to patients across the Middle East and North Africa region. The platform offers general medical consultations, health advice, and educational resources [15].

Advantages

- Provides online medical consultations and health education services.
- Features an intuitive and easy-to-use interface.
- Offers wide regional coverage across multiple MENA countries.
- Helps reduce hospital visits for minor health concerns.
- Includes limited features for tracking patient medical history.

Disadvantages

- Limited focus on specialized medical consultations.
- Does not provide advanced prioritization or triage for critical medical cases.
- Lacks structured reporting and efficient medical file management.
- The platform relies on external communication tools (such as Telegram) for patient-doctor communication instead of providing an integrated communication system.
- The platform does not provide a language translation option, which may create a barrier for Arabic-speaking users.
- Limited Tablet Responsiveness.



Figure 15 Mena Heal

2.3.3 Heal Gaza Overview

Heal Gaza is an online healthcare platform that allows patients to submit health-related inquiries and receive medical advice from available healthcare providers. The system primarily targets basic medical support for common and non-complex health issues [16].

Advantages

- Provides online health education and medical awareness resources.
- Simple and intuitive user interface suitable for users with basic internet access.
- Helps reduce unnecessary hospital visits for routine or minor cases.
- Encourages patient engagement and health awareness.
- Offers quick responses for general medical questions.
- Accessible on both desktop and mobile devices.

Disadvantages

- Limited availability of specialized physicians.
- Does not offer a structured system for uploading medical reports or conducting follow-ups.
- The platform provides various humanitarian services, but it is not specifically focused on managing digital medical consultations between patients and doctors.
- The platform does not provide a language translation option, which may create a barrier for Arabic-speaking users.



Figure 16 Heal gaza

2.4 Summary

In this chapter, we reviewed existing telemedicine platforms and similar projects that aim to provide online medical consultations in Gaza or similar contexts. While platforms like Gxza Health, MENA Health, and Heal Gaza offer basic online consultation and health resources, they have limitations in handling specialized medical cases, uploading medical reports, and providing secure and formal communication with physicians outside Gaza.

The GazaCare Connect system is proposed to address these gaps by offering secure, organized, and specialized telemedicine services. It supports uploading medical files, connecting patients with specialized physicians abroad, and ensuring high standards of privacy and usability. This comparison highlights the added value of the proposed system and its potential to improve access to advanced healthcare services in Gaza.

CHAPTER 3 RESEARCH METHODOLOGY

3.1 Introduction

The development of **Gaza Care Connect** requires a systematic and well-structured approach to address the unique healthcare challenges in the Gaza Strip, including the shortage of specialized medical professionals, limited resources, this chapter presents the research methodology adopted to design and develop a secure, reliable, and integrated telemedicine platform. Following modern software engineering practices, the methodology ensures that user requirements are effectively met while maintaining high standards of usability, performance, and data security.

Methodology Choice: Incremental Model

For this project, the Incremental Model was selected as the primary development methodology [17]. This model divides the system into small, manageable increments, where each increment goes through the phases of analysis, design, implementation, and testing before moving to the next stage [18].

Reasons for Choosing the Incremental Model for Gaza Care Connect:

Early Functional Prototype:

The model allows the delivery of a basic functional version of the system (such as user registration and consultation submission) early in the development lifecycle.

User Feedback Integration:

Each new increment is improved based on feedback from patients and physicians, ensuring the platform meets real healthcare needs.

Risk Management:

Developing the system in stages reduces technical risks and allows continuous testing of sensitive medical data security.

Feature Prioritization:

Core features such as secure medical report uploads and physician consultations are implemented before secondary functionalities.

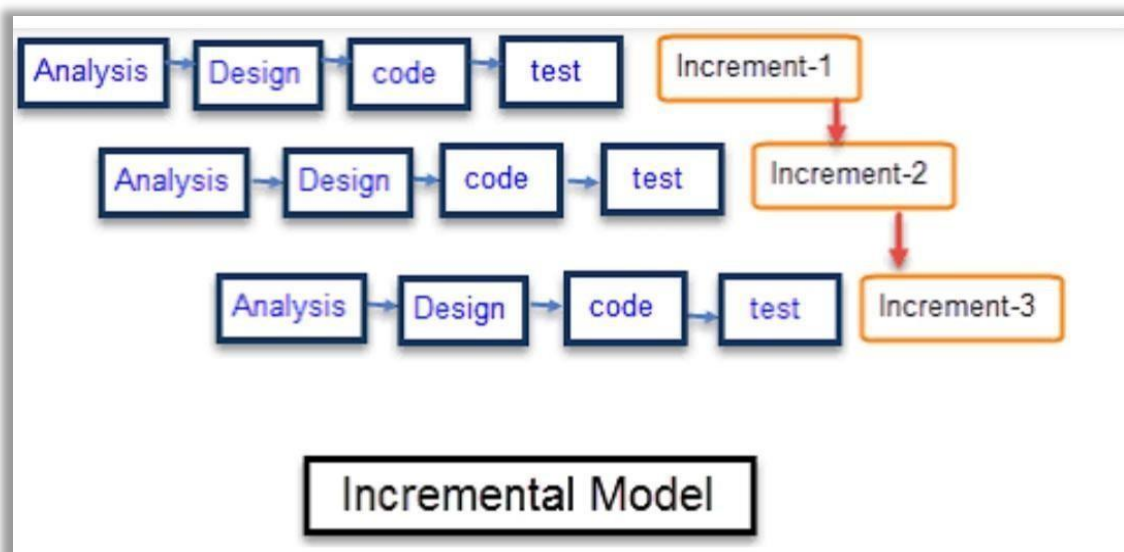


Figure 17 Incremental model

3.2 Development Phases

3.2.1 Analysis Phase

This phase focused on understanding the healthcare crisis in Gaza and identifying the needs of the two main user groups: patients inside Gaza and specialized physicians abroad.

Stakeholder Identification:

Defining the roles of patients (submitting medical cases), physicians (providing diagnoses and recommendations), and administrators (system management and monitoring).

Data Collection:

The analysis examined existing informal medical consultation channels such as social media, which lack privacy, reliability, and professional oversight.

User Personas:

Developing representative user profiles, such as a patient with a chronic illness seeking a second medical opinion from an external specialist.

3.2.2 Design Phase

The design phase emphasized creating a mobile-first, accessible, and user-friendly interface suitable for the Gaza context.

System Modeling:

Edraw Max and Draw.io were utilized to design system architecture diagrams, database schemas, and administrative workflows.

UI/UX Design:

Figma was used to design wireframes and interactive prototypes for key screens, including the Medical Profile and Consultation Dashboard.

Visual Elements:

Specialized screens were designed for entering health data (such as height, weight, and chronic diseases) and securely uploading medical files.

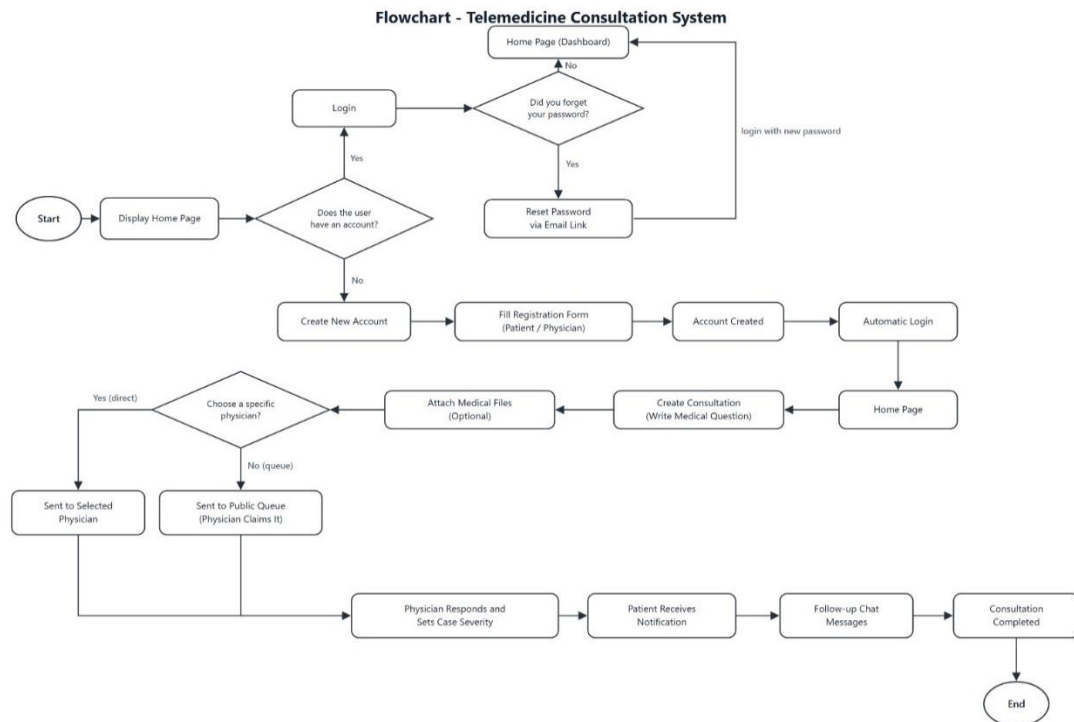


Figure 18 Workflow Diagram

3.2.3 Implementation / Coding Phase

The system was developed using a modern, scalable, and secure technology stack:

Front-End:

Developed using React.js and Next.js to ensure high performance and responsiveness, with

TypeScript for enhanced code reliability.

Back-End:

Implemented using the Laravel (PHP) framework, providing built-in security mechanisms for authentication, authorization, and data protection.

Database:

MySQL was used to store structured data, including patient profiles, medical records, and consultation history.

3.2.4 Testing Phase

Continuous testing was conducted to ensure system reliability under Gaza's infrastructure constraints.

Unit and Integration Testing:

Ensuring individual features (such as file uploads) function correctly and integrate seamlessly with the database.

Usability Testing:

Evaluating the ease of use for patients during key processes like submitting consultations.

Security Testing:

Verifying that all medical data is encrypted using SSL/HTTPS and accessible only to authorized users.

3.3 Survey Analysis Results

This section presents the findings of a statistical analysis conducted to identify the key healthcare challenges faced by patients in the Gaza Strip. The collected data reflects the urgent need for a secure and reliable telemedicine platform such as Gaza Care Connect.

A structured questionnaire was distributed to 65 participants, including patients suffering from chronic or specialized medical conditions and their primary caregivers.

3.3.1 Specialized Medical Needs Distribution

This initial statistical measure aims to identify the most common medical consultations that patients find difficult to access within the Gaza Strip due to limited local resources and referral restriction.

Survey Question:

Which simple medical consultation do you find most difficult to access within Gaza?

Results:

- **General Dermatology Consultation: 30%**
- **Common Cold/Flu Consultation: 25%**
- **Minor Gastrointestinal Consultation: 20%**

- **Mild Headache & Minor Pain Consultation: 25%**

Analysis:

The results indicate that dermatology and consultations for common cold/flu are the most requested and hardest to access, followed by minor gastrointestinal issues and mild headaches. These findings highlight that patients primarily need access to simple and quick medical consultations through remote telemedicine platforms.

System Implication:

Based on this result, specialist-to-patient teleconsultation was identified as a core functional requirement in the Gaza Care Connect system.

Specialized Medical Needs Distribution



Figure 19 Specialized Medical Needs Distribution

3.3.2 Reliability of Current Medical Communication Channels

This statistical measure examines the current methods used by patients to obtain second medical opinions from outside Gaza, focusing on reliability and data safety.

Survey Question:

What method do you currently use to obtain a second medical opinion from outside Gaza?

Results:

- **Unofficial Facebook and WhatsApp groups: 52%**
- **Personal or family connections: 28%**

- **Paid international medical platforms: 10%**
- **No available communication method: 10%**

Analysis:

The heavy reliance on informal social media platforms exposes patients to risks related to medical misinformation, lack of professional accountability, and severe breaches of data privacy.

System Implication:

Based on this finding, a verified and structured medical consultation environment was established as a primary system requirement.

Reliability of Current Channels

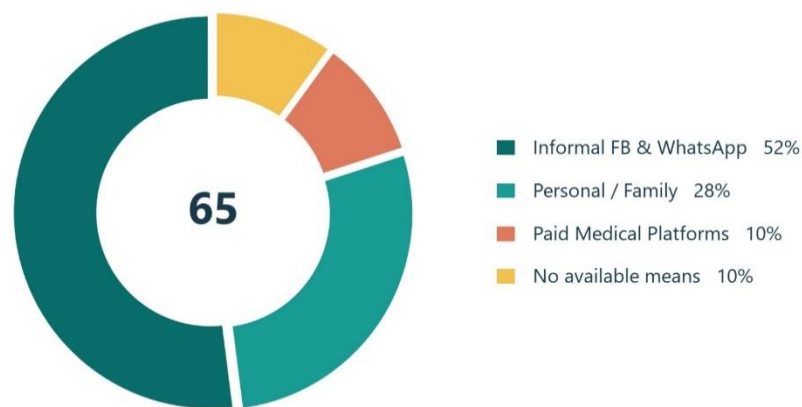


Figure 20 Reliability of Current Medical Communication Channels

3.3.3 Data Security and Privacy Awareness

Given the sensitive nature of medical records, this section evaluates user awareness and concern regarding data encryption and confidentiality.

Survey Question:

How important is it for you that your medical reports (lab tests, radiology images, diagnoses) are securely encrypted and kept confidential?

Results:

- **Very important: 94%**
- **Important: 4%**

- **Not important:** 2%

Analysis:

The overwhelming emphasis on data security reflects high user awareness of privacy risks. This validates the system’s architectural decision to prioritize secure backend technologies and encrypted data handling mechanisms.

System Implication:

Consequently, **secure encrypted file storage and transmission** were defined as mandatory non-functional system requirements, implemented using secure backend frameworks such as **Laravel**.

Trust in Medical Data Security & Privacy

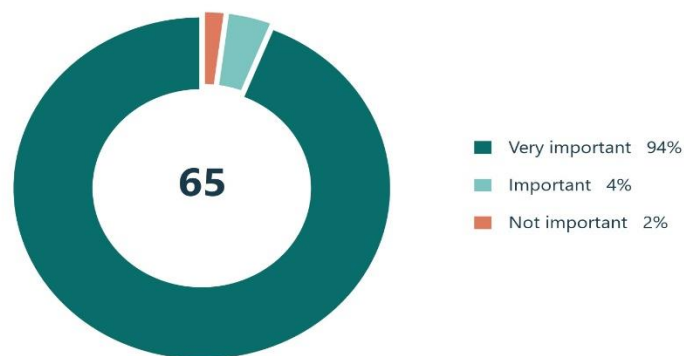


Figure 21 Data Security and Privacy Awareness

3.3.4 Technical Barriers to Platform Adoption

This subsection identifies potential technical obstacles that may hinder the adoption of an online medical consultation platform within Gaza’s constrained infrastructure.

Survey Question:

What is the primary technical barrier that may prevent you from using an online medical consultation platform?

Results:

- **Poor internet connectivity:** 45%
- **Difficulty uploading large medical files:** 35%
- **Unclear or complex user interface:** 20%

Analysis:

The results highlight infrastructure-related limitations as the most significant challenge, emphasizing the need for performance-optimized solutions and simplified user experiences.

System Implication:

In response, **lightweight frontend technologies**, such as **React** and **Next.js**, were selected to ensure high performance and usability under low-bandwidth conditions.

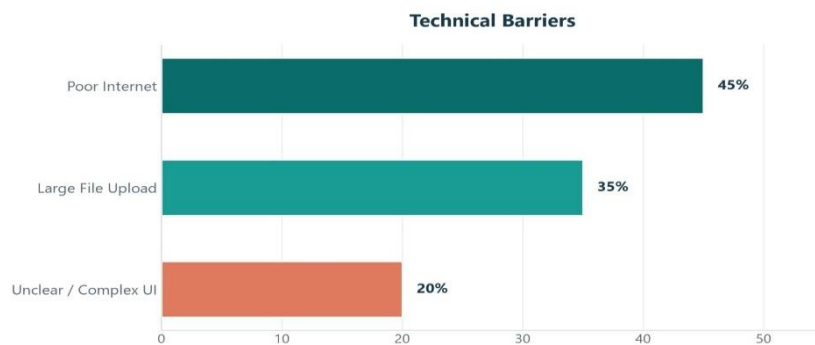


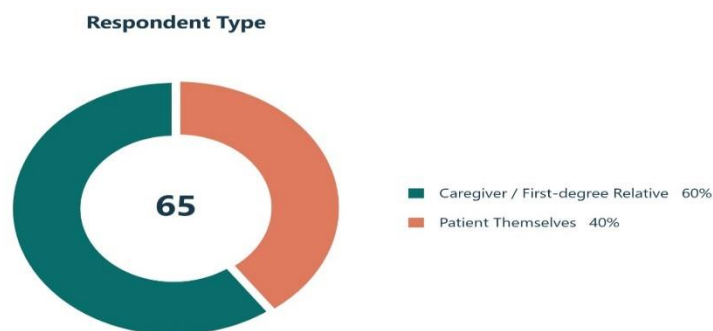
Figure 22 Technical Barriers to Platform Adoption

3.3.5 Demographic Considerations

Analysis of participant demographics revealed that approximately **60% of respondents were primary caregivers or first-degree relatives** of patients. This reflects the cultural and practical reality in Gaza, where caregivers often handle digital and administrative tasks on behalf of elderly patients or children.

System Implication:

Accordingly, the platform was designed to support **caregiver-managed patient profiles**, allowing authorized relatives to upload reports and communicate with specialists securely.



3.3.6 Summary of Analytical Findings

The survey results function as a diagnostic framework that translates healthcare challenges into measurable system requirements. Each statistical outcome directly influenced the design decisions and functional specifications of **Gaza Care Connect**, ensuring that the platform addresses real, context-specific medical needs rather than theoretical assumptions.

In this regard, the statistical analysis serves as a foundational pillar for system design, bridging the gap between healthcare challenges and technological solutions in a constrained environment.

3.4 Functional Requirements

The functional requirements define the core capabilities of the Gaza Care Connect platform:

3.4.1 Shared Functional Unit

RQ-01: Login

The system shall allow patients, physicians, and administrators to authenticate securely using email and password and redirect each user to the role-specific dashboard.

RQ-02: Reset Password

The system shall allow patients and physicians to reset a forgotten password through a secure email-based reset link.

RQ-03: Change Password

The system shall allow authenticated patients, physicians, and administrators to change their current password from account settings.

RQ-04: Update Profile

The system shall allow patients to update medical/personal profile data and physicians to update specialty and credential information.

RQ-05: Receive Notifications

The system shall deliver in-app notifications to patients, physicians, and administrators for key events (e.g., verification outcomes, case assignment, responses, and follow-up messages).

RQ-06: Logout

The system shall allow authenticated users to end their session securely and revoke access to protected pages until re-authentication.

3.4.2 Patient Functional Unit**RQ-07: Register as Patient**

The system shall allow a new user to create a patient account by submitting personal registration data, provided the email is not already registered.

RQ-08: Enable Caregiver Mode

The system shall allow a patient to enable caregiver mode and specify the family relationship when submitting consultations on behalf of a family member.

RQ-09: Browse Physicians by Specialty

The system shall allow patients to browse verified physicians and filter them by specialty.

RQ-10: View Physician Details

The system shall allow patients to view the full profile and specialty details of a selected verified physician.

RQ-11: Submit Consultation

The system shall allow patients to submit asynchronous text-based medical consultations either to a shared physician queue or directly to a selected verified specialist.

RQ-12: Upload Medical Reports

The system shall allow patients to attach medical reports and test results as images or PDF files to a consultation, subject to allowed formats and size limits 10MB.

RQ-13: View Consultation Status and Physician Response

The system shall allow patients to view consultation status, physician recommendations, attachments, and related message threads.

RQ-14: Send Follow-up Message

The system shall allow patients to send follow-up messages within a consultation thread after the physician's first response.

3.4.3 Physician Functional Unit**RQ-15: Register as Physician**

The system shall allow a physician to register with specialty and credential information and create an account in a pending-verification state until admin approval.

RQ-16: View Consultation Requests

The system shall provide a physician dashboard showing queue cases, directly assigned cases, in-progress cases, and completed cases.

RQ-17: Claim Queue Consultation

The system shall allow a verified physician to claim a pending consultation from the shared queue and notify the patient of assignment.

RQ-18: View Patient Details

The system shall allow an assigned physician to review the patient's medical profile and related consultation information, including caregiver details when applicable.

RQ-19: Download / View Medical Reports

The system shall allow an assigned physician to view or download medical files attached to a consultation.

RQ-20: Send Medical Recommendation

The system shall allow a verified physician to submit a medical recommendation, classify case severity (Mild / Moderate / Critical), optionally mark the case as completed, and notify the patient.

RQ-21: Send Follow-up Message

The system shall allow physicians to continue follow-up communication through a secure text message thread after the initial recommendation.

RQ-22: View Previous Cases

The system shall allow physicians to browse previously completed cases from the dashboard.

3.4.4 Admin Functional Unit**RQ-23: Manage Users**

The system shall allow administrators to view system users (patients and physicians) and review account statuses.

RQ-24: Disable User Account

The system shall allow administrators to disable or re-enable user accounts to control platform access.

RQ-25: Approve / Reject Physician

The system shall allow administrators to review pending physician credentials and approve or reject verification requests, with notification of the outcome.

3.5 Non-Functional Requirements

Non-functional requirements ensure the quality and sustainability of the system:

Security:

Sensitive medical data is protected through HTTPS/SSL in deployment, authenticated API access [19], and role-based authorization across patient, physician, and admin functions.

Usability:

A simple, intuitive Arabic RTL interface is tailored to the local community and designed for clear navigation across patient, physician, and admin workflows.

Performance:

The platform prioritizes a lightweight interface and asynchronous consultation flow to remain usable under low-bandwidth internet conditions common in Gaza.

Reliability:

System stability is supported through structured data storage, controlled access workflows, and operational practices aimed at continuity of service and data protection.

3.6 Conclusion of Methodology

By adopting the Incremental Model, Gaza Care Connect evolved from an analysis of local healthcare access needs into a functional digital teleconsultation solution. Core capabilities—secure authentication, medical profiles, text-based consultations with file attachments, physician verification, response workflows, follow-up messaging, and in-app notifications—were delivered in successive increments, allowing continuous refinement of security, usability, and reliability. Ultimately, the platform serves as a

secure digital bridge that improves access to specialized medical advice for patients in Gaza.

CHAPTER 4 SYSTEM USE CASES

4.1 Shared Use Cases

Table 1: Login

Table 1 Login

Field	Details
Use Case ID	UC-01 / RQ-01
Use Case Name	Login
Actors	Patient, Physician, Admin
Description	Users log into the system using their email and password.
Precondition	The user already has an account and the account is not disabled.
Basic Flow	1. User opens the login page. 2. Enters email and password. 3. System verifies credentials. 4. User is redirected to the role-based dashboard.
Alternate Flow	a. Invalid credentials → Error message shown. b. Disabled account → Access denied. c. Forgot password → Redirect to Reset Password.
Postcondition	User is authenticated and logged into the system.

Table 2: Reset Password*Table 2 Reset Password*

Field	Details
Use Case ID	UC-02 / RQ-02
Use Case Name	Reset Password
Actors	Patient, Physician, Admin
Description	User resets a forgotten password via a registered email.
Precondition	The user has a registered email address.
Basic Flow	1. User clicks “Forgot Password”. 2. Enters email. 3. System sends a reset link. 4. User opens the link and sets a new password.
Alternate Flow	Email not found → Error message.
Postcondition	Password is successfully updated.

Table 3: Change Password*Table 3 Change Password*

Field	Details
Use Case ID	UC-03 / RQ-03
Use Case Name	Change Password
Actors	Patient, Physician, Admin
Description	Authenticated user changes their current password.
Precondition	User is logged in.

Field	Details
Basic Flow	1. User opens account password settings. 2. Enters current and new password. 3. System validates and updates the password.
Alternate Flow	Incorrect current password or invalid new password → Error message.
Postcondition	Password is updated successfully.

Table 4: Update Profile

Table 4 Update Profile

Field	Details
Use Case ID	UC-04 / RQ-04
Use Case Name	Update Profile
Actors	Patient, Physician
Description	User updates personal and role-related profile information (medical profile for patients; specialty/credentials for physicians).
Precondition	User is logged in.
Basic Flow	1. User opens profile settings. 2. Edits information. 3. Saves changes. 4. System validates and stores updates.
Alternate Flow	Invalid data → Error message.
Postcondition	Profile information is updated.

Table 5: Receive Notifications

Table 5 Receive Notifications

Field	Details
Use Case ID	UC-05 / RQ-05
Use Case Name	Receive Notifications
Actors	Patient, Physician, Admin
Description	User receives in-app notifications about system events (e.g., physician verification, consultation assignment, replies, follow-up messages).
Precondition	User is logged in and the account is active.
Basic Flow	1. A system event occurs. 2. System creates a notification. 3. User opens the notification center and reads it. 4. User may mark notifications as read.
Alternate Flow	No notifications available → Empty notification list.
Postcondition	User is informed of relevant system events.

Table 6: Logout

Table 6 Logout

Field	Details
Use Case ID	UC-06 / RQ-06
Use Case Name	Logout
Actors	Patient, Physician, Admin
Description	User logs out of the system.
Precondition	User is logged in.
Basic Flow	1. User clicks logout. 2. System ends the session/token.

Field	Details
Alternate Flow	Not Applicable
Postcondition	Session ended; user must log in again to access protected pages.

4.2 Patient Use Cases

Table 7: Register as Patient

Table 7 Register as Patient

Field	Details
Use Case ID	UC-07 / RQ-07
Use Case Name	Register as Patient
Actors	Patient
Description	The patient creates a new account by providing required personal information.
Precondition	The patient does not have an existing account with the same email.
Basic Flow	1. Patient opens registration page. 2. Selects patient role. 3. Enters personal information. 4. Submits the form. 5. System validates data and creates the account.
Alternate Flow	a. Invalid data → Error message. b. Email already exists → System asks user to log in.
Postcondition	A new patient account is created and stored in the system.

Table 8: Enable Caregiver Mode*Table 8 Enable Caregiver Mode*

Field	Details
Use Case ID	UC-08 / RQ-08
Use Case Name	Enable Caregiver Mode
Actors	Patient
Description	Patient enables caregiver mode and specifies the relationship when submitting consultations on behalf of a family member.
Precondition	Patient is logged in.
Basic Flow	1. Patient opens caregiver mode settings. 2. Enables caregiver mode and selects relationship type. 3. System saves the setting.
Alternate Flow	Invalid relationship data → Error message.
Postcondition	Caregiver mode is enabled on the patient account.

Table 9: Browse Physicians by Specialty*Table 9 Browse Physicians by Specialty*

Field	Details
Use Case ID	UC-09 / RQ-09
Use Case Name	Browse Physicians by Specialty
Actors	Patient

Field	Details
Description	Patient browses verified physicians and may filter them by medical specialty.
Precondition	Patient is logged in.
Basic Flow	1. Patient opens the physician's directory. 2. Optionally selects a specialty filter. 3. System displays matching verified physicians.
Alternate Flow	No verified physicians available for the selected specialty → Empty list message.
Postcondition	List of verified physicians is visible to the patient.

Table 10: View Physician Details

Table 10 View Physician Details

Field	Details
Use Case ID	UC-10 / RQ-10
Use Case Name	View Physician Details
Actors	Patient
Description	Patient views detailed information about a selected verified physician.
Precondition	Physicians list is displayed.
Basic Flow	1. Patient selects a physician. 2. System displays specialty and profile details.
Alternate Flow	Physician not found / not verified → Error or unavailable message.
Postcondition	Physician information is shown.

Table 11: Submit Consultation*Table 11 Submit Consultation*

Field	Details
Use Case ID	UC-11 / RQ-11
Use Case Name	Submit Consultation
Actors	Patient
Description	Patient submits an asynchronous text consultation either to the shared physician queue or directly to a selected verified physician.
Precondition	Patient is logged in.
Basic Flow	1. Patient opens “New Consultation”. 2. Writes the medical question (text). 3. Chooses assignment mode: queue or direct. 4. If direct, selects a verified physician. 5. Optionally attaches medical files(max 20 files). 6. Submits the consultation. 7. System stores the case as pending.
Alternate Flow	a. Invalid/short question → Error message. b. Direct mode without physician → Validation error. c. Selected physician unavailable → Error message.
Postcondition	A new consultation is created and awaiting physician handling.

Table 12: Upload Medical Reports*Table 12 Upload Medical Reports*

Field	Details
Use Case ID	UC-12 / RQ-12
Use Case Name	Upload Medical Reports
Actors	Patient

Field	Details
Description	Patient uploads medical reports (images or PDF) and attaches them to a consultation.
Precondition	Patient is logged in; consultation is being created or is still editable before physician reply.
Basic Flow	1. Patient selects upload option. 2. Chooses an image or PDF file. 3. System uploads and stores the file. 4. File is linked to the consultation.
Alternate Flow	Unsupported format, oversized file over than 10MB, or upload failure → Error message.
Postcondition	Medical report is available for the assigned/claiming physician.

Table 13: View Consultation Status and Physician Response

Table 13 View Consultation Status and Physician Response

Field	Details
Use Case ID	UC-13 / RQ-13
Use Case Name	View Consultation Status and Physician Response
Actors	Patient
Description	Patient opens a consultation to view its status, physician recommendation, case severity, and follow-up messages.
Precondition	Patient has at least one consultation; for response viewing, the physician has replied.
Basic Flow	1. Patient opens consultations list. 2. Selects a consultation. 3. System displays question, attachments, status, recommendation, and message thread (if any).
Alternate Flow	Consultation still pending with no reply → Only patient question/attachments are shown.

Field	Details
Postcondition	Patient is informed of consultation progress and medical advice when available.

Table 14: Send Follow-up Message

Table 14 Send Follow-up Message

Field	Details
Use Case ID	UC-14 / RQ-14
Use Case Name	Send Follow-up Message
Actors	Patient
Description	After the physician's first response, the patient sends follow-up text messages within the consultation thread.
Precondition	Physician has already responded to the consultation.
Basic Flow	1. Patient opens the consultation thread. 2. Writes a follow-up message. 3. Submits the message. 4. System stores it and notifies the physician.
Alternate Flow	Empty message or consultation not eligible for messaging → Error message.
Postcondition	Follow-up message is added to the consultation thread.

4.3 Physician Use Cases

Table 15: Register as Physician

Table 15 Register as Physician

Field	Details
Use Case ID	UC-15 / RQ-15

Field	Details
Use Case Name	Register as Physician
Actors	Physician
Description	Physician creates an account with specialty and credential information pending admin verification.
Precondition	Physician does not have an existing account with the same email.
Basic Flow	1. Physician opens registration page. 2. Selects physician role. 3. Enters personal data, specialty, and certificate information/files. 4. Submits the form. 5. System creates account with pending verification and notifies admins.
Alternate Flow	a. Invalid data → Error message. b. Email already exists → Ask user to log in.
Postcondition	Physician account is created and awaits admin approval before handling consultations.

Table 16: View Consultation Requests

Table 16 View Consultation Requests

Field	Details
Use Case ID	UC-16 / RQ-16
Use Case Name	View Consultation Requests
Actors	Physician
Description	Verified physician views queue cases, direct assignments, in-progress cases, and completed cases from the dashboard.
Precondition	Physician is logged in and verified.

Field	Details
Basic Flow	1. Physician opens the dashboard. 2. System displays consultation sections (queue / direct / in progress / completed).
Alternate Flow	No requests available → Empty sections. Unverified physician → Restricted from claiming/responding.
Postcondition	Consultation requests list is visible.

Table 17: Claim Queue Consultation

Table 17 Claim Queue Consultation

Field	Details
Use Case ID	UC-17 / RQ-17
Use Case Name	Claim Queue Consultation
Actors	Physician
Description	Verified physician claims a pending consultation from the shared queue.
Precondition	Physician is verified; a pending queue consultation exists.
Basic Flow	1. Physician opens the queue section. 2. Selects a consultation. 3. Claims the case. 4. System assigns the consultation to the physician and notifies the patient.
Alternate Flow	Case already claimed by another physician → Error/unavailable message.
Postcondition	Consultation is assigned to the claiming physician.

Table 18: View Patient Details*Table 18 View Patient Details*

Field	Details
Use Case ID	UC-18 / RQ-18
Use Case Name	View Patient Details
Actors	Physician
Description	Physician views the patient's profile and medical information related to a consultation.
Precondition	A consultation assigned to or claimed by the physician is selected.
Basic Flow	1. Physician opens the consultation details. 2. System displays patient personal/medical profile and caregiver info if enabled.
Alternate Flow	Not Applicable
Postcondition	Patient details are displayed.

Table 19: Download / View Medical Reports*Table 19 Download / View Medical Reports*

Field	Details
Use Case ID	UC-19 / RQ-19
Use Case Name	Download / View Medical Reports
Actors	Physician
Description	Physician views or downloads medical files attached to a consultation.

Field	Details
Precondition	Patient has uploaded reports to the consultation.
Basic Flow	1. Physician opens the attachments section. 2. Views or downloads files.
Alternate Flow	No reports available → Empty attachments list.
Postcondition	Reports are accessed by the physician.

Table 20: Send Medical Recommendation

Table 20 Send Medical Recommendation

Field	Details
Use Case ID	UC-20 / RQ-20
Use Case Name	Send Medical Recommendation
Actors	Physician
Description	Physician sends a medical recommendation, assigns case severity, and optionally marks the consultation as completed.
Precondition	Consultation is assigned to the verified physician.
Basic Flow	1. Physician writes the recommendation. 2. Selects case severity (mild / moderate / critical). 3. Optionally marks the case as completed. 4. Submits the response. 5. System notifies the patient.
Alternate Flow	Missing recommendation or severity → Validation error.
Postcondition	Recommendation is sent; consultation is updated (pending review or completed).

Table 21: Send Follow-up Message

Table 21 Send Follow-up Message

Field	Details
Use Case ID	UC-21 / RQ-21
Use Case Name	Send Follow-up Message
Actors	Physician
Description	Physician continues the consultation through the secure text message thread after the initial response.
Precondition	Physician has already responded to the consultation.
Basic Flow	1. Physician opens the consultation thread. 2. Writes a message. 3. Submits it. 4. System stores the message and notifies the patient.
Alternate Flow	Empty message → Error message.
Postcondition	Follow-up message is added to the consultation thread.

Table 22: View Previous Cases

Table 22 View Previous Cases

Field	Details
Use Case ID	UC-22 / RQ-22
Use Case Name	View Previous Cases
Actors	Physician
Description	Physician browses completed/previous consultations from the dashboard.
Precondition	Physician is logged in.

Field	Details
Basic Flow	1. Physician opens the completed/previous cases section. 2. Browses the list and opens case details if needed.
Alternate Flow	No previous cases → Empty list.
Postcondition	Previous cases are displayed.

4.4 Admin Use Cases

Table 23: Manage Users

Table 23 Manage Users

Field	Details
Use Case ID	UC-23 / RQ-23
Use Case Name	Manage Users
Actors	Admin
Description	Admin views system users (patients and physicians) and reviews account status.
Precondition	Admin is logged in.
Basic Flow	1. Admin opens user's management page. 2. Views the list of users and roles.
Alternate Flow	No users found → Empty list.
Postcondition	User list is reviewed.

Table 24: Disable User Account*Table 24 Disable User Account*

Field	Details
Use Case ID	UC-24 / RQ-24
Use Case Name	Disable User Account
Actors	Admin
Description	Admin disables a user account to prevent further access when necessary.
Precondition	Target user account exists.
Basic Flow	1. Admin selects a user. 2. Chooses disable/enable action. 3. Confirms the action. 4. System updates account status.
Alternate Flow	Action cancelled by admin.
Postcondition	Account is disabled (or re-enabled); disabled users cannot access protected features.

Table 25: Approve / Reject Physician*Table 25 Approve / Reject Physician*

Field	Details
Use Case ID	UC-25 / RQ-25
Use Case Name	Approve / Reject Physician
Actors	Admin
Description	Admin reviews pending physician credentials and approves or rejects verification.

Field	Details
Precondition	At least one physician registration is pending verification.
Basic Flow	1. Admin opens pending physicians page. 2. Reviews specialty and certificate information. 3. Approves or rejects the physician. 4. System updates verification status and notifies the physician.
Alternate Flow	No pending physicians → Empty list.
Postcondition	Physician is verified (can handle consultations) or rejected with the reason for rejection and the ability to resubmit a verification request.

CHAPTER 5 SYSTEM MODELING AND DESIGN

5.1 GazaCare Connect Activity Diagram

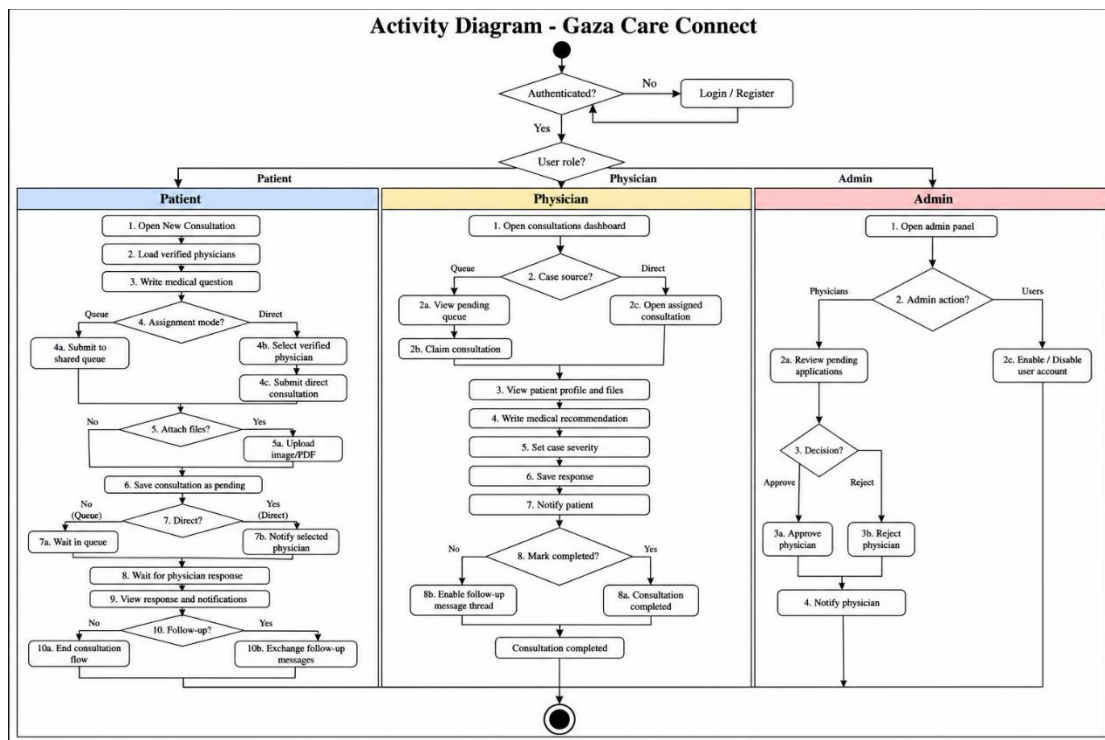


Figure 24 Activity Diagram — Main Consultation Flow

This activity diagram describes the main workflow of Gaza Care Connect, from authentication to consultation handling across Patient, Physician, and Admin roles.

Process Flow:

- The process starts when a user opens the system.
- If the user is not authenticated, they must Login or Register.
- After authentication, the flow branches according to the user role.

Patient Flow:

- The patient opens New Consultation.
- The system loads verified physicians.
- The patient writes the medical question.
- The patient chooses assignment mode: Queue or Direct.
- Optionally, the patient uploads medical files (image/PDF).
- The consultation is saved as pending.
- If direct mode is selected, the selected physician is notified; otherwise, the case waits in the shared queue.
- The patient waits for the physician response, then can view the reply and send follow-up messages if needed.

Physician Flow:

- The physician opens the consultations dashboard.
- The physician either claims a case from the queue or opens a direct assignment.
- The physician views the patient profile and attached files.
- The physician writes a medical recommendation and sets case severity.
- The response is saved and the patient is notified.
- The physician may mark the consultation as completed, or leave it open for follow-up messaging.

Admin Flow:

- The admin opens the admin panel.
- The admin reviews pending physician applications and approves or rejects them.
- The admin may also enable/disable user accounts.

This activity diagram shows the complete decision-based workflow of Gaza Care Connect and clarifies how consultations move from submission to response and follow-up.

5.2 GazaCare Connect Use Case Diagram

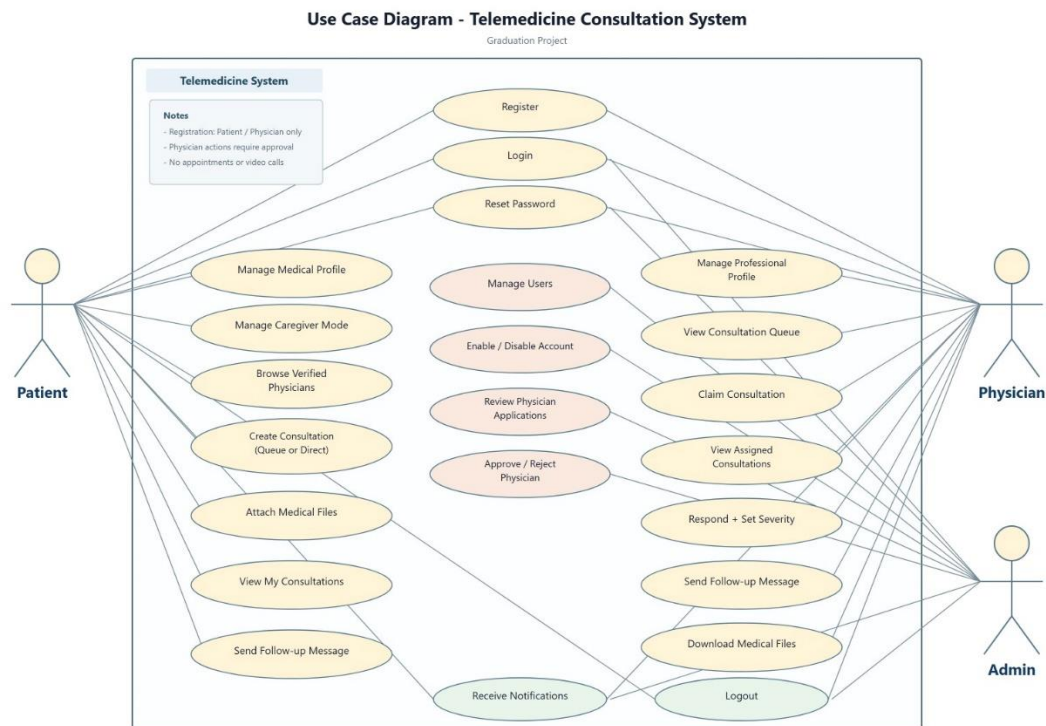


Figure 25 Use Case Diagram

The following Use Case Diagram represents the Unified Modeling Language (UML) modeling of the GazaCare Connect telemedicine system. The diagram illustrates the interactions between the primary system actors and the core functionalities provided by the platform.

The system includes three major actors:

- Admin
- Patient
- Physician

5.2.1 Admin Actor Responsibilities

The Admin is responsible for managing system operations and user control, including:

- Managing users
- Enabling or disabling user accounts
- Reviewing physician applications
- Approving or rejecting physician accounts

5.2.2 Physician Actor Responsibilities

The Physician (Specialist) interacts with the system to provide remote medical advice, including:

- Registering a new physician account (requires admin approval)
- Logging into the system
- Resetting password
- Managing professional profile
- Viewing the consultation queue
- Claiming consultations
- Viewing assigned consultations
- Responding to consultations and setting case severity
- Downloading medical files
- Sending follow-up messages
- Receiving notifications
- Logging out of the system

5.2.3 Patient Actor Responsibilities:

The Patient interacts with the system to access telemedicine services, including:

- Registering a new account

- Logging into the system
- Resetting password
- Managing medical profile
- Managing caregiver mode
- Browsing verified physicians
- Creating consultations (queue or direct)
- Attaching medical files
- Viewing own consultations
- Sending follow-up messages
- Receiving notifications
- Logging out of the system

The diagram also includes shared use cases such as Register, Login, Reset Password, Receive Notifications, and Logout. Physician registration requires admin approval before the physician can handle consultations. The system does not include appointments or video calls.

This use case diagram provides a high-level functional overview of system behavior and user interaction in the Telemedicine System (Gaza Care Connect).

5.3 GazaCare Connect Sequence Diagram

5.3.1 User Register

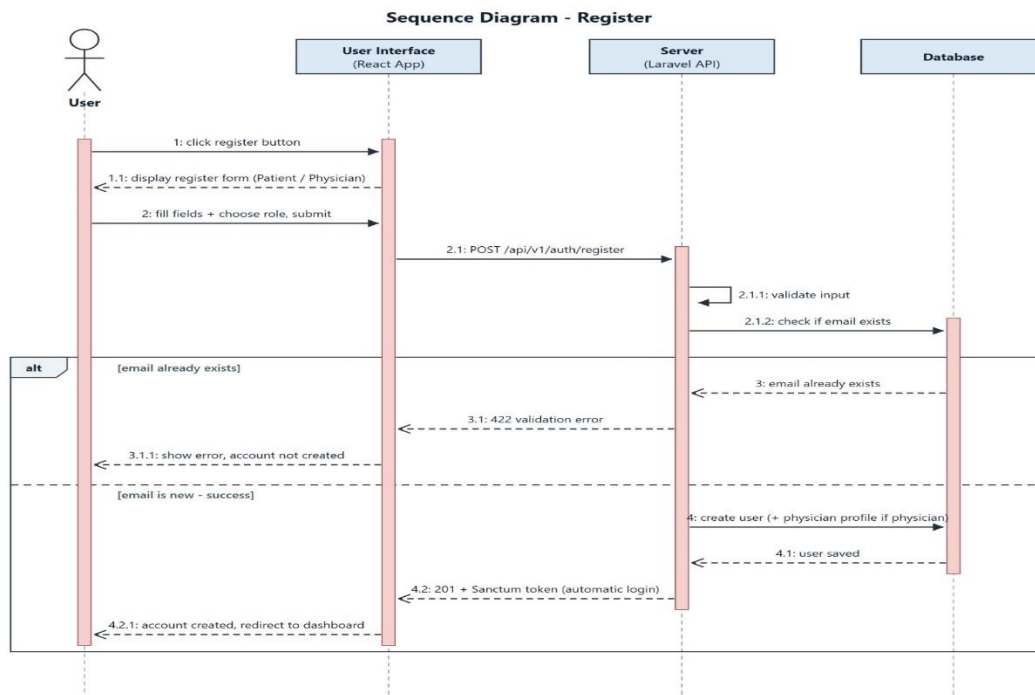


Figure 26 User Register

This sequence diagram describes the registration process in Gaza Care Connect.

Process Flow:

- The User clicks the register button.
- The User Interface (React App) displays the registration form (Patient or Physician).
- The user fills in the required fields, selects a role, and submits.
- The UI sends a POST `/api/v1/auth/register` request to the Server (Laravel API).
- The server validates the input and checks with the Database whether the email already exists.

Alternative Fragment:

- **Failed Registration [email already exists]**

1. The database indicates that the email already exists.
2. The server returns a 422 validation error.
3. The UI displays an error message and the account is not created.

- **Successful Registration [email is new]**

1. The server creates the user in the database (and a physician profile if the Physician role is selected).
2. The database confirms that the user is saved.
3. The server returns 201 with a Sanctum token (automatic login).
4. The UI confirms that the account was created and redirects the user to the dashboard.

This sequence ensures validated registration and secure authentication before granting system access.

5.3.2 User Login

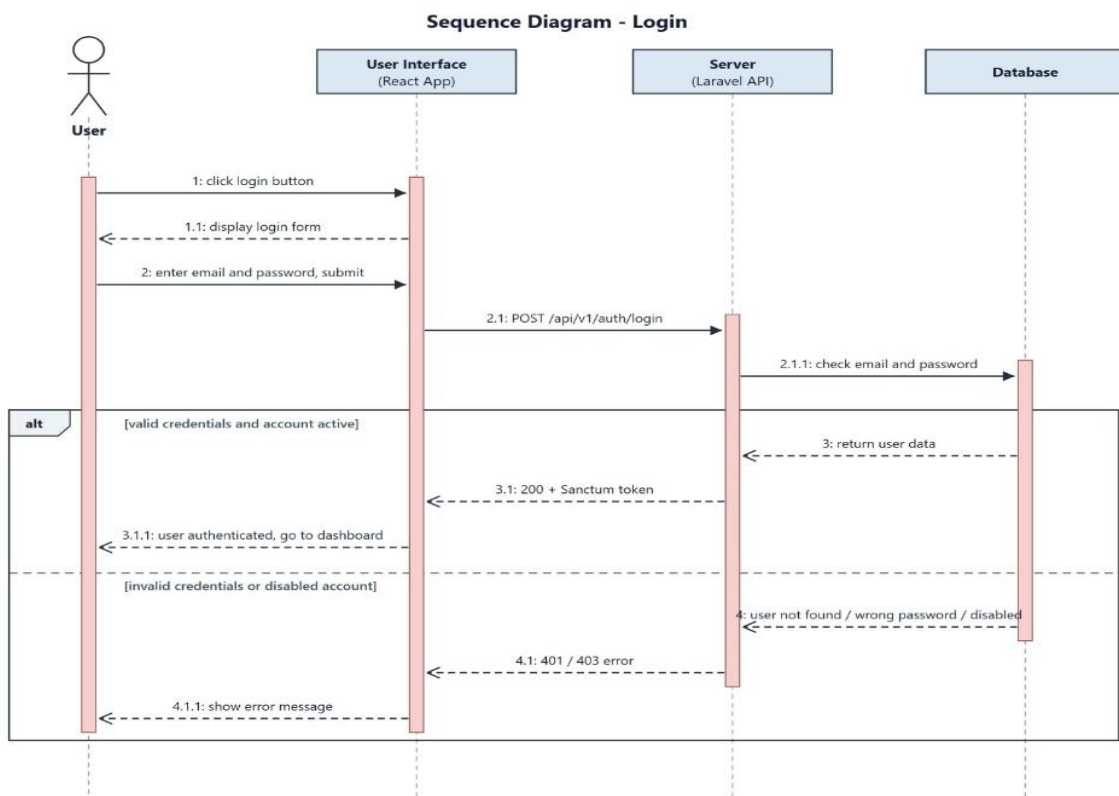


Figure 27 User Login

This sequence diagram describes the authentication process in Gaza Care Connect.

Process Flow:

- The User clicks the login button.
- The User Interface (React App) displays the login form.
- The user enters email and password, then submits.
- The UI sends a POST /api/v1/auth/login request to the Server (Laravel API).
- The server verifies the email and password with the Database.

Alternative Fragment:

- **Successful Login [valid credentials and account active]**
 1. The database returns the user data.
 2. The server authenticates the user and returns 200 with a Sanctum token.
 3. The UI redirects the authenticated user to the dashboard.
- **Failed Login [invalid credentials or disabled account]**
 1. The database indicates that the user was not found, the password is wrong, or the account is disabled.
 2. The server returns a 401 / 403 error.
 3. The UI displays an error message to the user.

This sequence ensures secure authentication through the Laravel API and Sanctum token before granting system access.

5.3.3 Reset Password

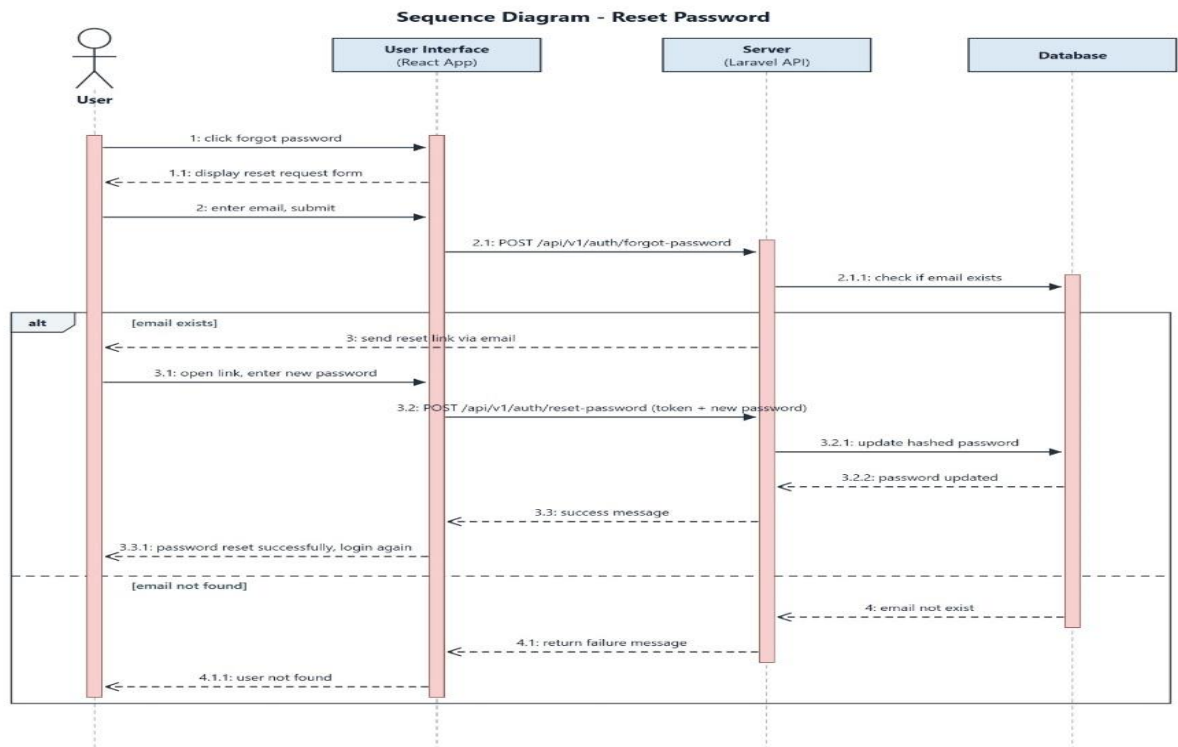


Figure 28 Reset Password

This sequence diagram describes the password reset process in Gaza Care Connect.

Process Flow:

- The User clicks “Forgot Password”.
- The User Interface (React App) displays the reset request form.
- The user enters their email and submits.
- The UI sends a POST `/api/v1/auth/forgot-password` request to the Server (Laravel API).
- The server checks with the Database whether the email exists.

Alternative Fragment:

- **Successful Reset [email exists]**
 1. The server sends a reset link to the user’s email.
 2. The user opens the link and enters a new password.
 3. The UI sends a POST `/api/v1/auth/reset-password` request with the token and new password.
 4. The server updates the hashed password in the database.

5. The database confirms the update.
 6. The server returns a success message, and the UI informs the user that the password was reset successfully and to log in again.
- **Failed Reset [email not found]**
 1. The database indicates that the email does not exist.
 2. The server returns a failure message.
 3. The UI displays a “user not found” error.

This sequence ensures a secure password recovery flow before allowing the user to access the system again.

5.3.4 Edit User Profile

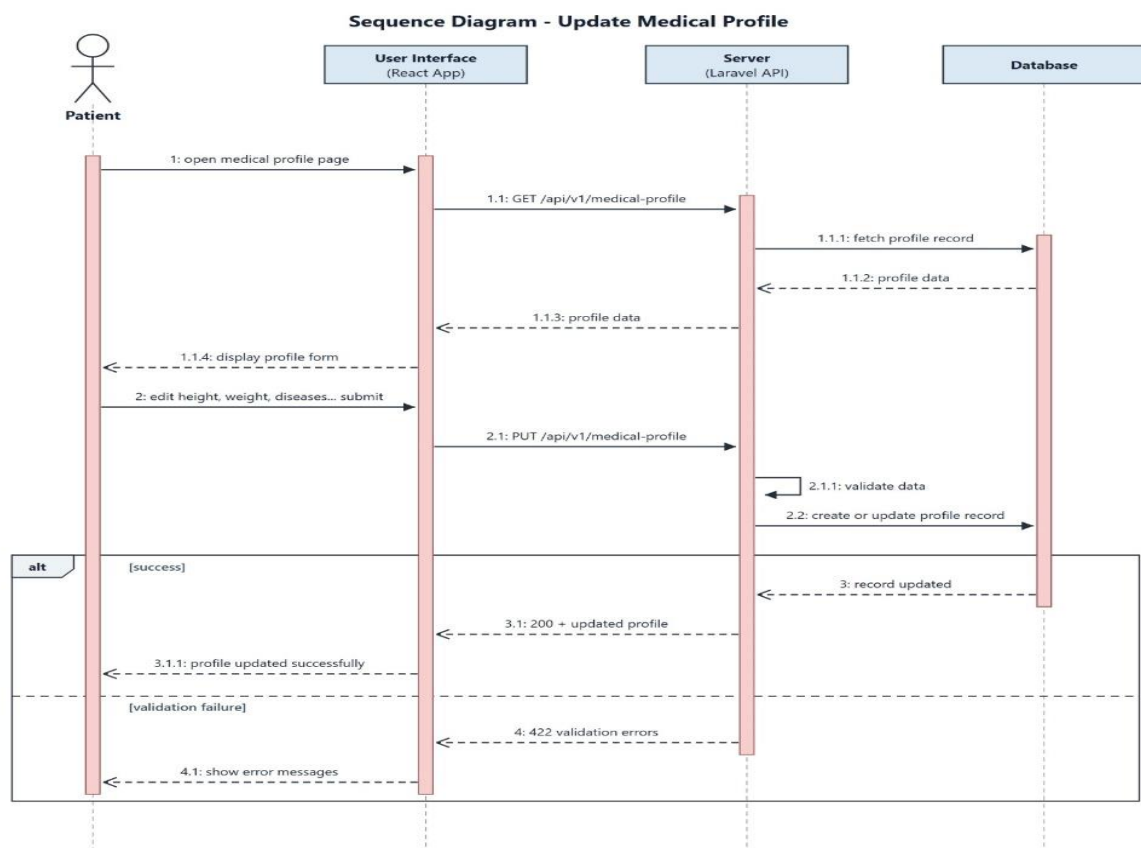


Figure 29 Edit User Profile

This sequence diagram describes the process of updating a patient’s medical profile in Gaza Care Connect.

Process Flow:

- The Patient opens the medical profile page.

- The User Interface (React App) sends a GET /api/v1/medical-profile request to the Server (Laravel API).
- The server fetches the profile record from the Database and returns the profile data.
- The UI displays the profile form to the patient.
- The patient edits fields such as height, weight, and diseases, then submits.
- The UI sends a PUT /api/v1/medical-profile request to the server.
- The server validates the data and creates or updates the profile record in the database.

Alternative Fragment:

- **Successful Update [success]**
 1. The database confirms that the record was updated.
 2. The server returns 200 with the updated profile.
 3. The UI displays a “profile updated successfully” message.
- **Failed Update [validation failure]**
 1. The server returns a 422 validation error.
 2. The UI displays error messages to the patient.

This sequence ensures that medical profile data is loaded, validated, and securely updated before being saved in the system.

5.3.5 Create Consultation

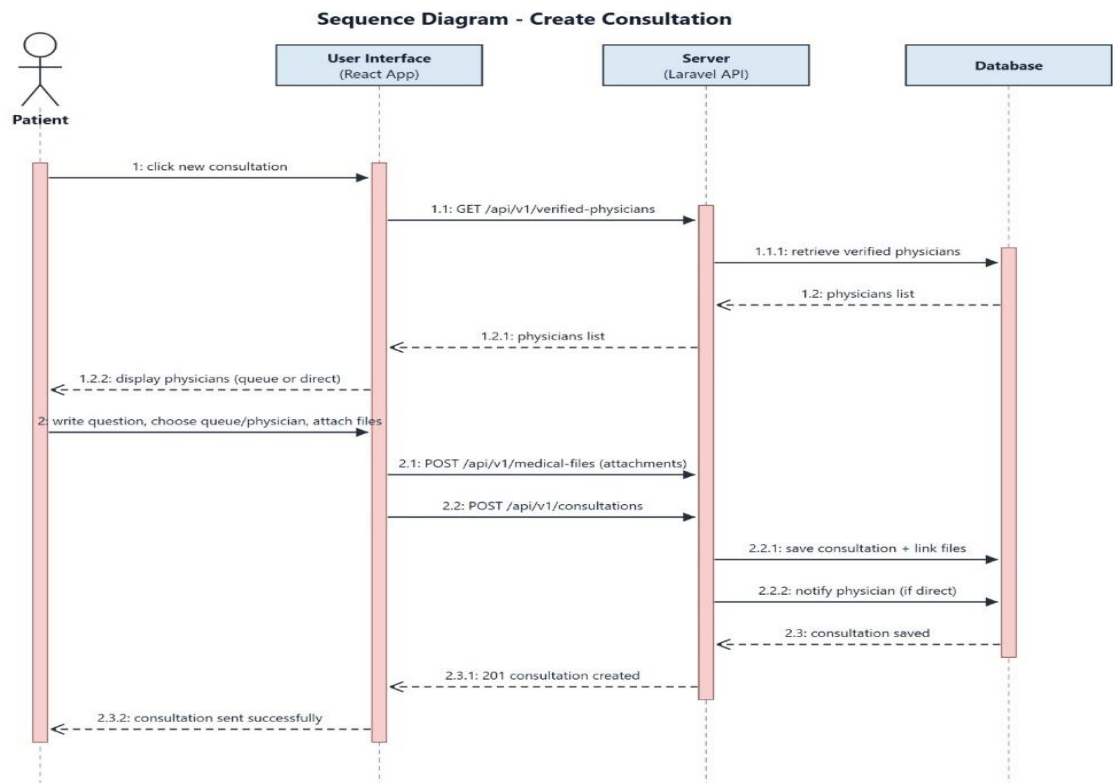


Figure 30 Create Consultation

This sequence diagram describes the process of creating a medical consultation in Gaza Care Connect.

Process Flow:

- The Patient clicks “New Consultation”.
- The User Interface (React App) sends a GET `/api/v1/verified-physicians` request to the Server (Laravel API).
- The server retrieves the verified physicians list from the Database and returns it to the UI.
- The UI displays the physicians and assignment options (queue or direct).
- The patient writes the question, chooses queue or a specific physician, and attaches files if needed.
- The UI uploads attachments using POST `/api/v1/medical-files`, then submits the consultation using POST `/api/v1/consultations`.

- The server saves the consultation, links the files, and notifies the physician if the assignment mode is direct.
- The database confirms that the consultation is saved.
- The server returns 201 consultation created.
- The UI displays a “consultation sent successfully” message to the patient.

This sequence ensures that consultations are created with the correct assignment mode, attachments, and physician notification when required.

5.3.6 Physician Responds

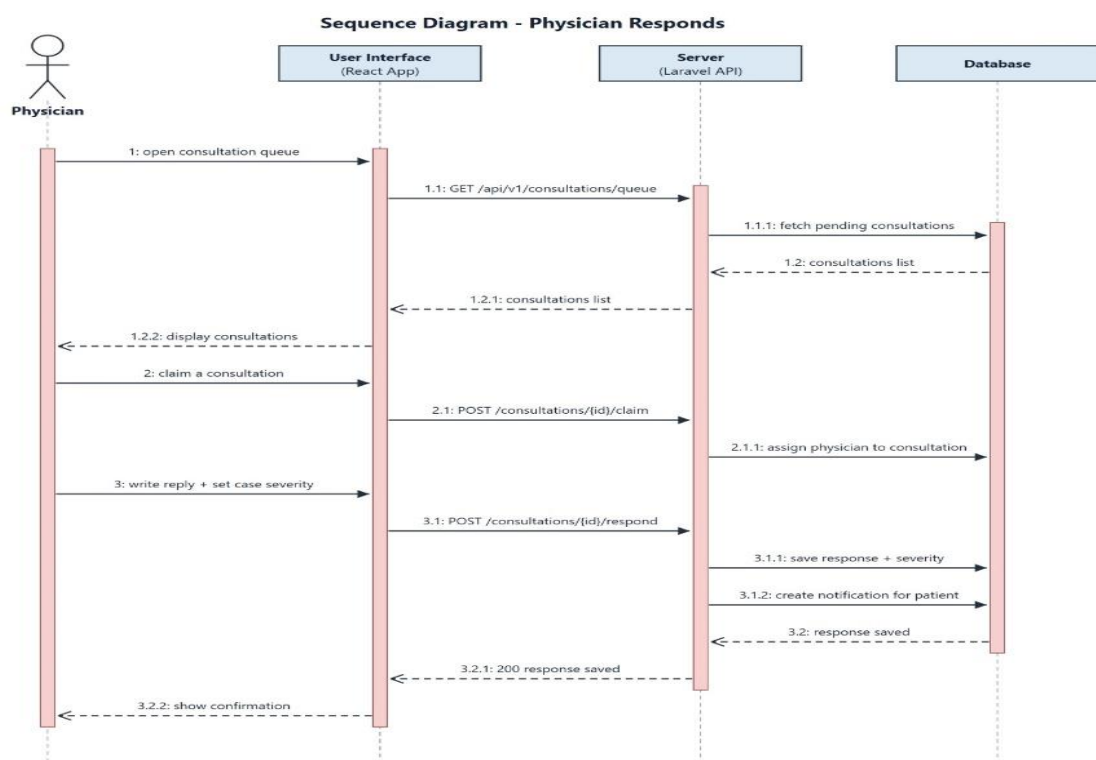


Figure 31 Physician Responds

This sequence diagram describes the process of a physician claiming and responding to a consultation in Gaza Care Connect.

Process Flow:

- The Physician opens the consultation queue.
- The User Interface (React App) sends a GET /api/v1/consultations/queue request to the Server (Laravel API).

- The server fetches pending consultations from the Database and returns the list to the UI.
- The UI displays the consultations list to the physician.
- The physician claims a consultation.
- The UI sends a POST `/api/v1/consultations/{id}/claim` request, and the server assigns the physician to the consultation in the database.
- The physician writes a reply and sets the case severity.
- The UI sends a POST `/api/v1/consultations/{id}/respond` request to the server.
- The server saves the response and severity, then creates a notification for the patient.
- The database confirms that the response is saved.
- The server returns 200 response saved, and the UI shows a confirmation message to the physician.

This sequence ensures that consultations are claimed, answered with severity classification, and that patients are notified after the physician responds.

5.3.7 Logout

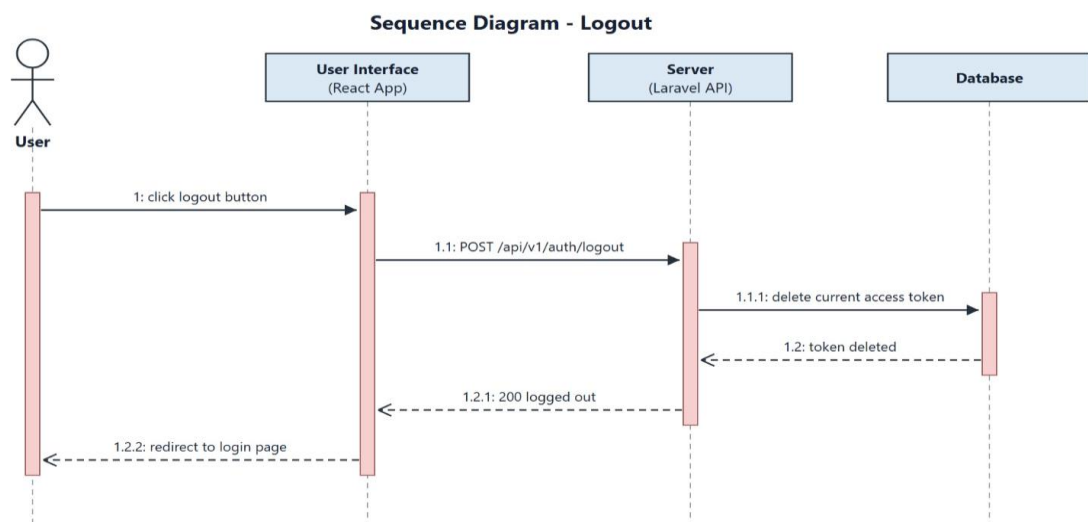


Figure 32 Logout

This sequence diagram describes the logout process in Gaza Care Connect.

Process Flow:

- The User clicks the logout button.
- The User Interface (React App) sends a POST /api/v1/auth/logout request to the Server (Laravel API).
- The server deletes the current access token from the Database.
- The database confirms that the token has been deleted.
- The server returns 200 logged out.
- The UI redirects the user to the login page.

This sequence ensures that the user session is securely terminated by invalidating the Sanctum access token before returning to the login screen.

5.4 GazaCare Connect Class Diagram

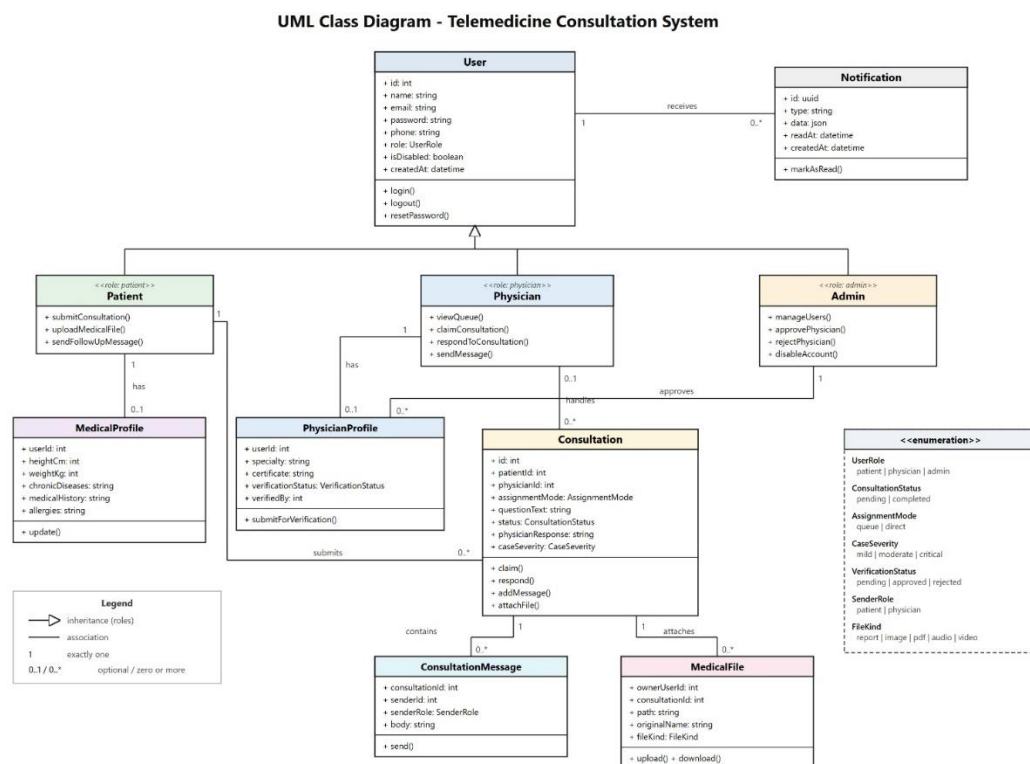


Figure 33 Class Diagram

Class Diagram Description

The UML Class Diagram represents the structural design of the Gaza Care Connect system.

Core Structure:

The User class serves as the base class for:

- Patient
- Physician
- Admin

Common attributes include:

- id
- name
- email
- password
- phone
- role
- isDisabled
- createdAt

Each subclass extends the base class with specialized methods:

- Patient: submitConsultation(), uploadMedicalFile(), sendFollowUpMessage()
- Physician: viewQueue(), claimConsultation(), respondToConsultation(), sendMessage()
- Admin: manageUsers(), approvePhysician(), rejectPhysician(), disableAccount()

Key Domain Classes:

- MedicalProfile → Stores the patient's health data (height, weight, chronic diseases, history, allergies).
- PhysicianProfile → Stores specialty, certificate, and verification status.
- Consultation → Represents a medical consultation with assignment mode, question, status, response, and case severity.
- ConsultationMessage → Represents follow-up text communication within a consultation.
- MedicalFile → Stores uploaded medical documents linked to a consultation.

- Notification → Represents system notifications received by users.

Enumerations for:

- UserRole
- ConsultationStatus
- AssignmentMode
- CaseSeverity
- VerificationStatus
- SenderRole
- FileKind

The class diagram provides a complete structural blueprint for system implementation and ensures proper object-oriented design principles.

5.5 Chapter Conclusion

This chapter presented a comprehensive modeling and system design analysis of Gaza Care Connect using UML diagrams.

The use case diagrams clarified actor–system interactions for Patient, Physician, and Admin roles. The sequence diagrams illustrated dynamic communication flows between the React App, Laravel API, and Database for key processes such as login, registration, password reset, medical profile updates, consultation creation, physician response, and logout. The activity diagram described the decision-based workflow of the consultation lifecycle across roles. The class diagram defined the structural architecture of the platform, including users, profiles, consultations, messages, medical files, and notifications.

Together, these models establish a well-organized development framework and contribute to building a secure, structured, and scalable telemedicine system that meets user requirements effectively.

CHAPTER 6 SYSTEM IMPLEMENTATION

6.1 Development Environment

The GazaCare Connect system was developed using a modern web development environment that supports security, scalability, and maintainability. The development process relied on a combination of front-end and back-end technologies, supported by a relational database management system and professional development tools.

Table 26 Software Tools, Frameworks, and Technologies

Component	Technology	Purpose
Front-End	React.js	Building interactive and responsive user interfaces.
Front-End Framework	Next.js	Routing, rendering, and improving application performance.
Programming Language	TypeScript	Improving code quality through static typing.
Back-End	Laravel	Implementing business logic and RESTful APIs.
Server-Side Language	PHP	Processing requests and communicating with the database.
Database	MySQL	Storing users, consultations, medical records, and attachments.
Database Tool	phpMyAdmin	Database administration and management.

The front-end application was implemented using React.js and Next.js to provide a responsive and user-friendly interface for patients, doctors, and administrators. TypeScript was adopted to improve code reliability and maintainability throughout the project.

The back-end was developed using the Laravel framework, which provides authentication, routing, middleware, validation, and secure API development. PHP was used as the primary server-side programming language to process application requests and manage communication with the database.

MySQL was selected as the relational database management system because of its reliability, structured data storage capabilities, and compatibility with Laravel. phpMyAdmin was used to simplify database administration, table management, and data inspection during development and testing.

Visual Studio Code served as the primary integrated development environment (IDE), Git was utilized to track source code changes and facilitate collaborative development among team members.

6.2 System Architecture

The GazaCare Connect system follows a three-tier architecture that separates the presentation layer, application layer, and data layer. This architectural approach improves system maintainability, scalability, and security by separating user interaction, business logic, and data management into independent components.

The architecture consists of the following layers:

Front-End Layer

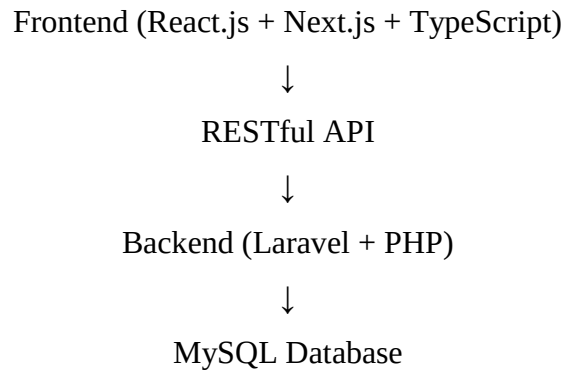
The presentation layer was developed using React.js and Next.js with TypeScript to provide a responsive, user-friendly, and interactive web interface. This layer is responsible for displaying system pages, collecting user input, validating forms, and communicating with the back-end through RESTful API requests.

Back-End Layer

The application layer was implemented using the Laravel framework with PHP. It contains the core business logic of the system, including user authentication, authorization, consultation management, medical report handling, profile management, and role-based access control. Laravel also manages communication between the client application and the database while ensuring secure request processing.

Database Layer

The data layer was implemented using MySQL. It stores all system information, including user accounts, patient medical profiles, doctor profiles, consultations, uploaded medical reports, and recommendations. The database was designed using relational tables to ensure data consistency and efficient retrieval.



In this architecture, users interact with the web interface through their browsers. The front-end sends requests to the Laravel REST API, which processes the requests, applies the required business logic, interacts with the MySQL database, and returns the appropriate responses to the client application. This architecture ensures secure communication, organized code structure, and efficient data management throughout the system.

6.3 Database Implementation

The GazaCare Connect database was implemented using MySQL as the relational database management system (RDBMS). It was designed to provide secure data storage, efficient retrieval, and well-structured organization of user and medical information. The database structure was developed based on the system requirements identified during the analysis and design phases while ensuring data integrity through the use of primary and foreign key relationships.

The database schema was implemented using Laravel migration files, allowing database tables and relationships to be created, modified, and maintained in a structured and consistent manner. This approach simplifies database deployment, supports version control, and ensures compatibility across different development environments.

Unlike traditional telemedicine systems that maintain separate tables for patients and physicians, GazaCare Connect stores all user accounts in a single users table. User roles, including patient, physician, and administrator, are distinguished through a dedicated role attribute. Information specific to physicians and patients is stored in separate profile tables that are linked to the users table through foreign key relationships. Likewise, consultation responses are stored directly within the

consultations table and through the consultation_messages table, eliminating the need for a separate recommendations table.

The database consists of several interconnected tables, each responsible for storing a specific category of information. User-related tables manage authentication and account information, while other tables store medical profiles, consultation records, uploaded medical files, physician responses, and system notifications. The main database tables and their functions are summarized in Table.

Table 27 Main Database Tables

Table	Description
users	Stores authentication and account information for all system users, including role, phone number, account status, and optional caregiver mode settings.
physician_profiles	Stores physicians' professional information, including specialty, certification details, verification status, and administrator verification records.
medical_profiles	Stores each patient's medical profile, including medical history, allergies, chronic diseases, current medications, height, and weight. Each patient has one associated medical profile.
consultations	Stores consultation requests submitted by patients, including the assigned physician, assignment mode (queue or direct), consultation status, physician response, case severity, and response timestamps.
consultation_messages	Stores the conversation thread exchanged between patients and physicians during each consultation.
medical_files	Stores uploaded medical reports, images, PDF documents, and other attachments associated with a user and, when applicable, a consultation.
notifications	Stores in-application notifications related to consultations, account activities, and other system events.
personal_access_tokens	Stores API authentication tokens generated and managed by Laravel Sanctum for secure user authentication.

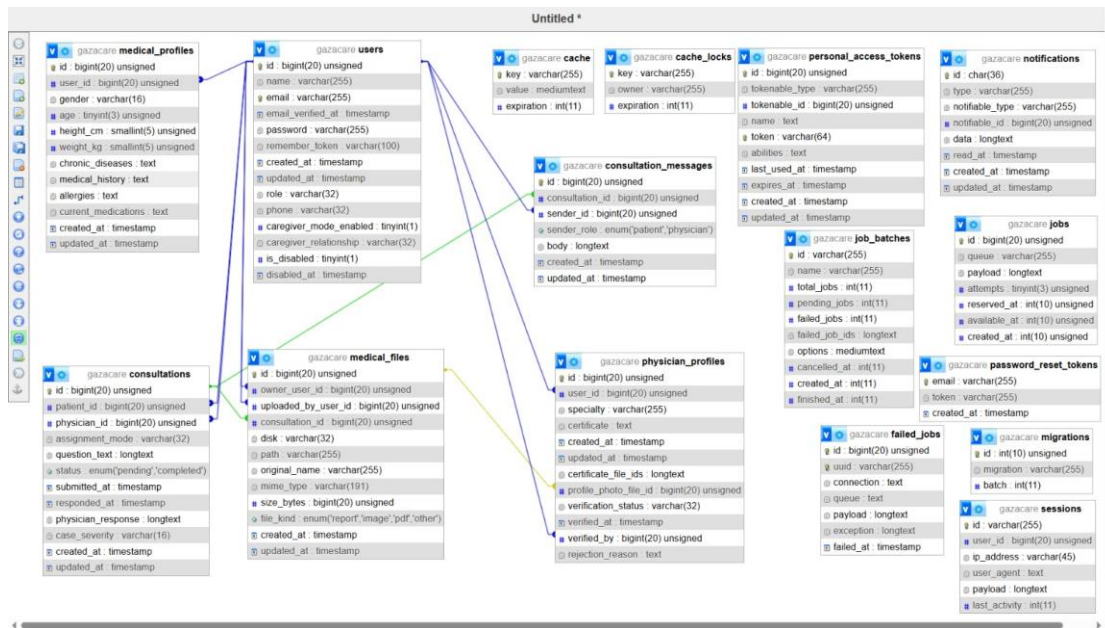


Figure 34 Database Schema

6.4 System Modules

The GazaCare Connect platform was implemented as a collection of integrated functional modules, each responsible for a specific set of operations while collaborating to provide a complete asynchronous telemedicine platform. This modular architecture enhances system maintainability, simplifies future development and feature expansion, and promotes a clear separation of responsibilities among authentication, consultation management, physician verification, notifications, and administrative functions.

6.4.1 Authentication Module

The Authentication Module manages user registration, authentication, password management, and secure access to the platform. Laravel Sanctum was utilized to authenticate API requests using personal access tokens and to protect restricted system resources. User passwords are securely hashed before being stored in the database, ensuring the confidentiality of authentication credentials.

The module supports patient and physician registration, user login and logout, password reset via email, and authenticated password changes. It also implements role-based authentication and authorization, ensuring that patients, physicians, and administrators are granted access only to the functionalities associated with their roles. In addition, disabled accounts are prevented from logging into the system or accessing protected resources.

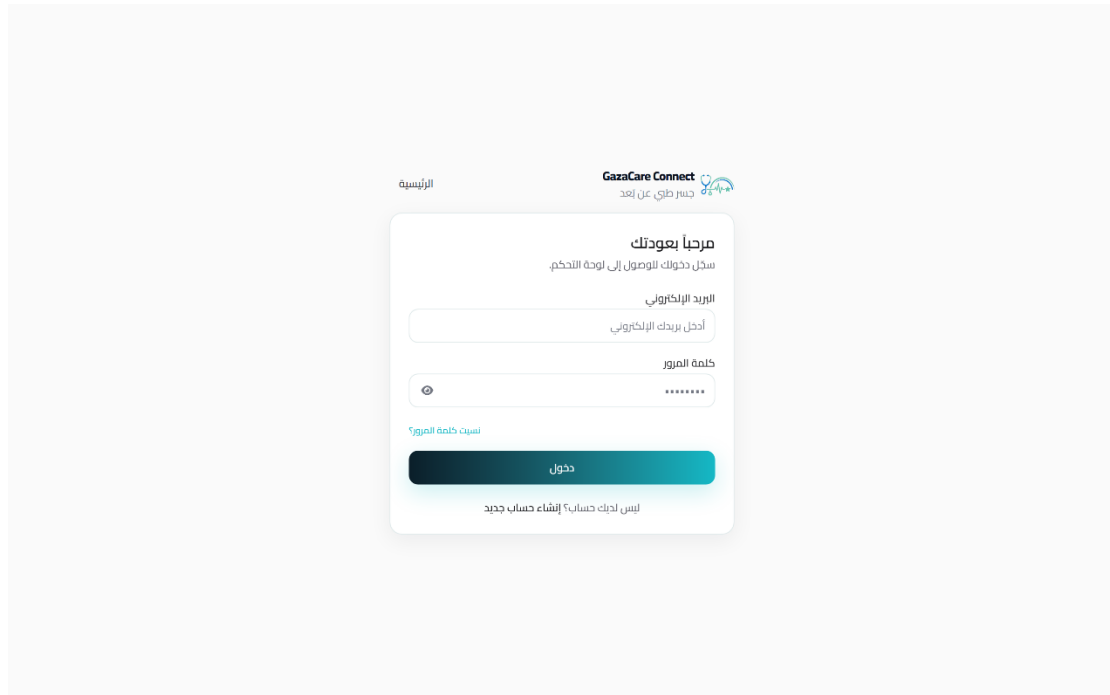


Figure 35 Login page of GazaCare Connect

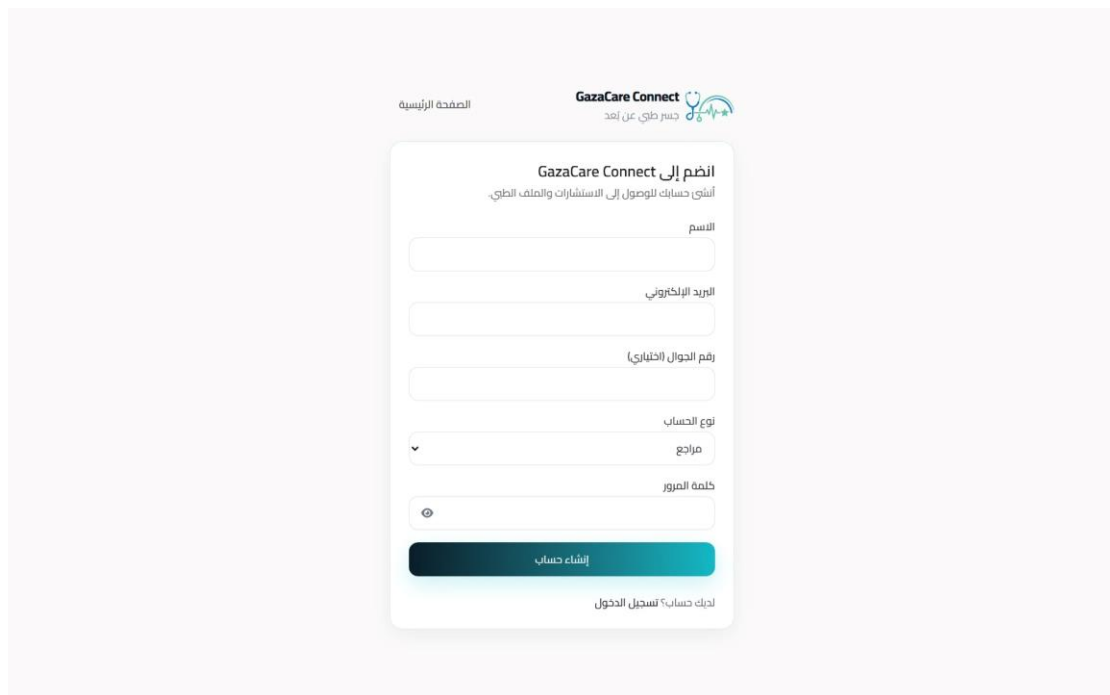


Figure 36 Patient registration form

الصفحة الرئيسية

GazaCare Connect

جسر طبي عن بعد

انضم إلى GazaCare Connect

أنتهي حسابك للوصول إلى الاستشارات والملف الطبي. حساب الطبيب يخضع لمراجعة الإدارة قبل استكمال الحالات.

الاسم

البريد الإلكتروني

رقم الجوال (اختياري)

نوع الحساب

طبيب

أدخل تخصصك ومؤهلك. ثم أرفق صورتك الشخصية ذاتيات هوية وشهادتك (PDF أو صورة) ساردها الإدارة قبل تفعيل حسابك.

الصورة الشخصية (إلزامية)

صورة واضحة لوجهك — استخدم ذاتيات هوية عند مراجعة طلب التوثيق.

اختيار الصورة

صورة وجّه واضحة (JPG أو PNG)

التخصص

مثالي: طب الأطفال، جراحة عامة، قلب...

الشهادة / المؤهل

مثالي: بكالوريوس طب وجراحة، بورد/مؤهل، جامعة...

أكتبوا بشكل مختصر وواضح (الحد الأقصى 5000 حرف).

مرفقات الشهادة

اختيار ملفات الشهادة

أرفق شهادة التخرج أو البورد (PDF أو صورة). يمكنك اختيار أكثر من ملف.

كلمة المرور

إنشاء حساب

لديك حساب؟ تسجل الدخول

Figure 37 Physician registration form with credential upload

78

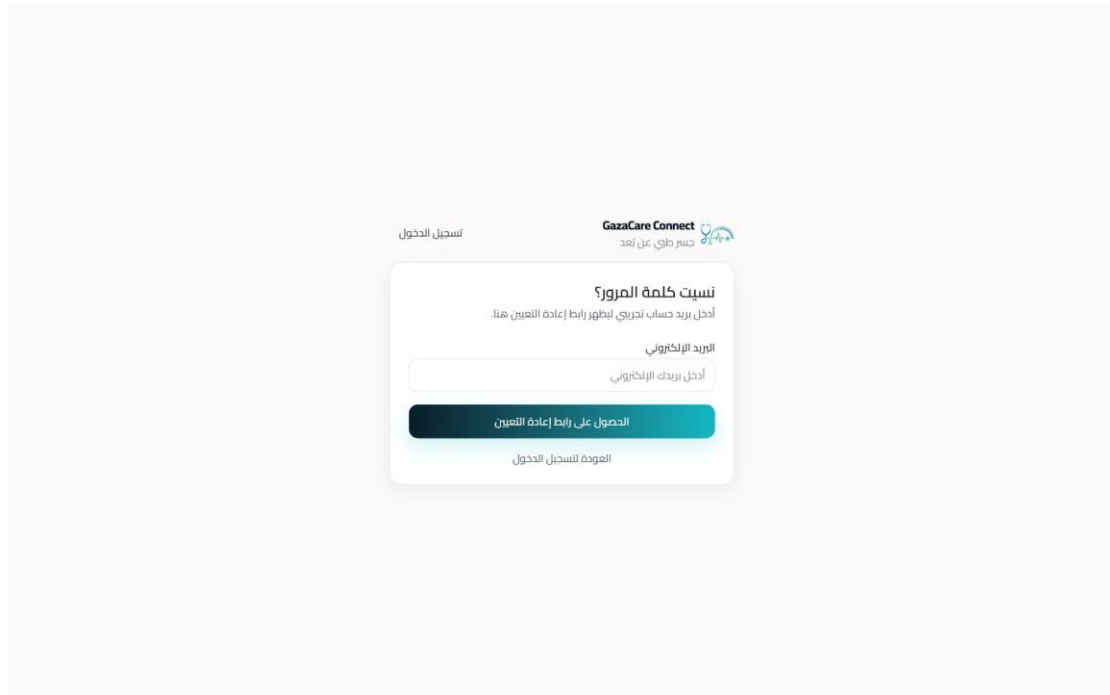


Figure 38 Password reset request page

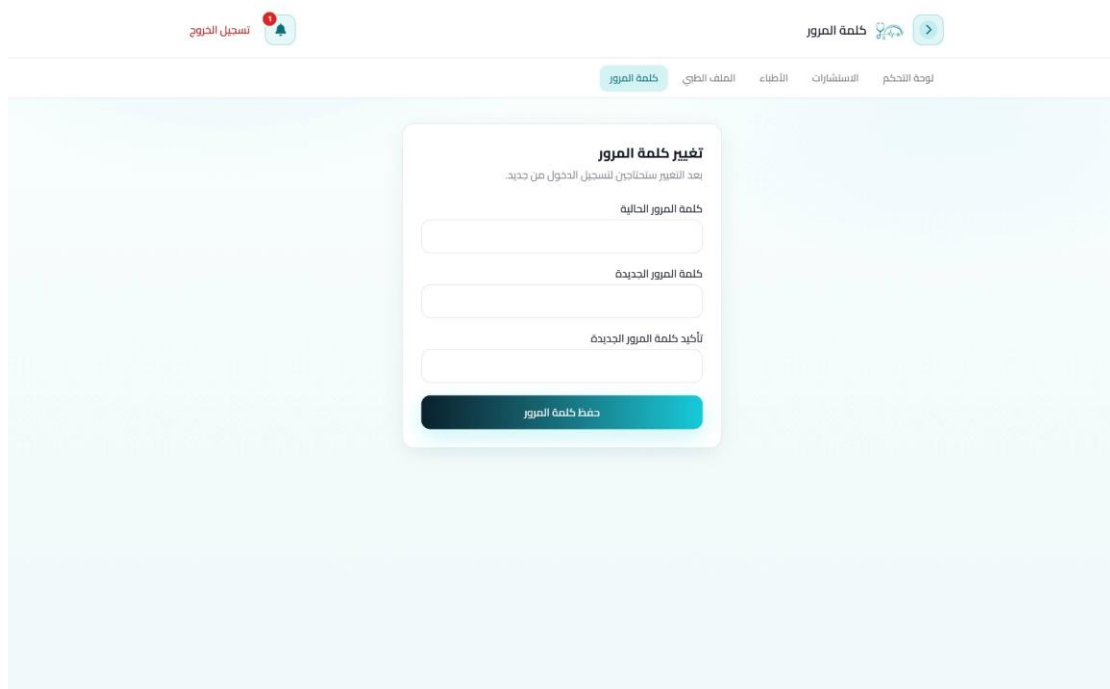


Figure 39 Authenticated password change page

6.4.2 User Profile Module

The User Profile Module enables users to manage their personal and role-specific information. Patients can create and update their personal and medical profiles, including height, weight, chronic diseases, allergies, medical history, and current

medications. This medical information is stored securely in the database and is available to physicians during consultations to support informed medical decision-making.

Physicians maintain professional profiles containing their medical specialty and supporting credential documents. Newly registered physician accounts remain in a pending state until verified by an administrator, and unverified physicians are restricted from handling patient consultations.

Figure 40 Patient medical profile management

Figure 41 Physician medical profile management

6.4.3 Caregiver Mode Module

The Caregiver Mode Module allows patients to indicate that a consultation is being submitted on behalf of a family member. Users can enable caregiver mode and specify the relationship type. This information is displayed to physicians and administrators during consultation review, providing additional context regarding the individual receiving medical advice.

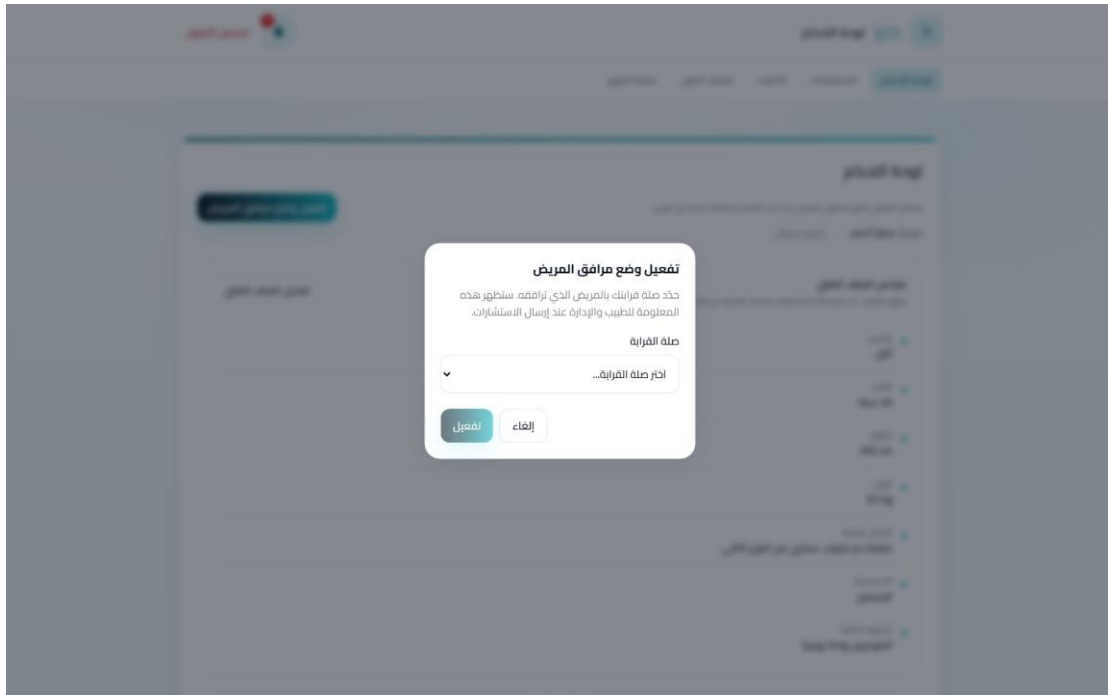


Figure 42 Caregiver mode activation dialog

6.4.4 Physician Directory Module

The Physician Directory Module enables patients to browse verified physicians and filter them according to medical specialty. Patients can view physician profiles, including specialty information and verification status, before selecting a physician and submitting a direct consultation.

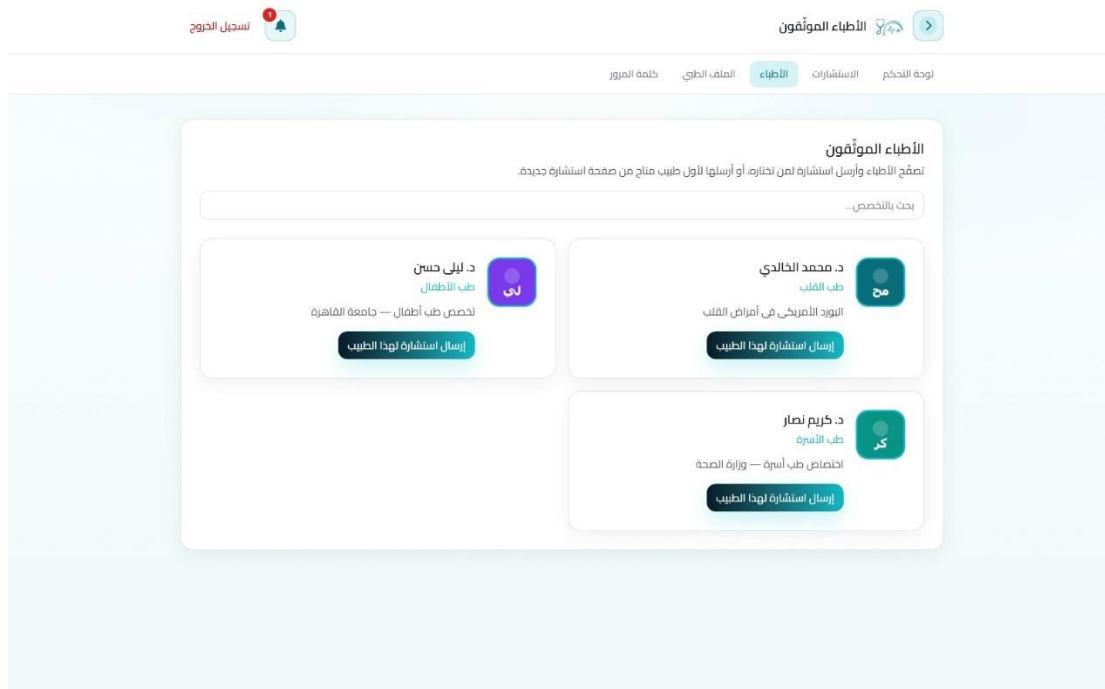


Figure 43 Browsing verified physicians by specialty

6.4.5 Consultation Module

The Consultation Module represents the core functionality of the GazaCare Connect platform. It enables patients to submit asynchronous text-based medical consultations either to a shared consultation queue or directly to a selected verified physician. Consultation records include the patient's medical question, consultation status, assignment method, uploaded attachments, physician responses, and timestamps.

Verified physicians can access consultation requests assigned directly to them or claim pending consultations from the shared queue. They can review patient medical information and attached files, provide medical recommendations together with a case severity level (mild, moderate, or critical), and update consultation status throughout the consultation lifecycle. Patients can monitor consultation progress, review physician recommendations, and access consultation details through their dashboards.

إرسال استشارة

اختر طريقة الإرسال، اكتب سؤالك، وأرفق الملفات عند الحاجة.

طريقة الإرسال

1

أرسلها لأول طبيب محتاج

تبقى بانتظار أول طبيب محتاج لاستلامها.

طبيب محدد

لرسل مباشرة إلى الطبيب الذي تختاره.

ملفك الطبي

2

إرفق مع الاستشارة راجع البيانات قبل الإرسال.

تعدل الملف

الجنس

أنثى

العمر

45 سنة

الطول

165 cm

الوزن

62 kg

أمراض مزمنة

ضغط دم خفيف، سكري من النوع الثاني

التاريخ الحالي

تشخيص السكري منذ 2019. مرافقة ضغط الدم في مركز الشفاء.

الخصائص

البيانات

الادوية الحالية

أملوديبين 5mg يوميا

سؤال الاستشارة

3

صف أعراضك أو اكتب سؤالك الطبي.

مثال: صداع منذ أسبوعين مع دوخة عند الوقوف، هل أحتاج فحوصات؟

نحل 10 أحرف على الأقل (0/100)

مرفقات (اختياري)

4

تقرير، أشعة، أو تحليل — PDF أو صورة.

اختيار ملفات

لم تختار ملفات بعد

Figure 44 Submitting a consultation to the shared queue

إرسال استشارة

اختر طريقة الإرسال، اكتب سؤالك، وأرفق الملفات عند الحاجة.

طريقة الإرسال

أرسلها لأول طبيب متاح
لنلقي بالنظر أول طبيب متاح نساعدك.

اختر الطبيب

د. محمد الخالدي — طب القلب

تصفح الأطباء المؤهلين

سأُرسل إلى د. د محمد الخالدي

ملفك الطبي

إزعم مع الاستشارة راجع البيانات قبل الإرسال.

تعديل الملف

الجنس
أنثى

العمر
45 سنة

الطول
165 cm

الوزن
62 kg

أمراض مزمنة
ضغط دم خفيف، سكري من النوع الثاني

التاريخ الطبي
تشخيص السكري منذ 2019، مراقبة ضغط الدم في مركز الشفاء.

الجسمانية
البينسالين

الحساسية
ألمودجيبين 5mg يومياً

سؤال الاستشارة

صف أعراضك أو اكتب سؤالك الطبي

مثال: صداع منذ أسبوعين مع دوخة عند الوقوف، هل أحتاج محوصات؟

أدخل 10 أحرف على الأقل (0/10)

مرفقات (اختياري)

تقرير أشعة، أو تحليل — PDF أو صورة

اختر ملفات

لم أختار ملفات بعد

إرسال الاستشارة

إلغاء

Figure 45 Submitting a direct consultation to a selected physician

سجل الاستشارات

كل استشارتك هنا

مؤلفة لطيب محدد

أرسلتها لطبيب معين وهي بانتظار ردم.

ارسالت 29 جولائی 2026

#8 قيد الانتظار طبيب محدد

نص الاستشارة

أرسلت هذه الاستشارة مباشرة للدكتور محمد. أعالي من ألام في المفاصل منذ أسبوع.

عرض التفاصيل ←

بانتظار رد الطيب

بانتظار طبيب — قيد الانتظار

ما زالت بالنظر أن يستلمها أول طبيب متاح.

ارسالت 28 يوليو 2026

قيد الانتظار #7

نص الاستشارة

أشعر بخفقان في القلب أحياناً بعد المجهود. هل هذا طبيعي مع ضغط الدم؟

عرض التفاصيل ←

بانتظار رد الطيب

மீகை

الطبيب رد عليها.

ارسالت 23 يوليو 2026

#2 **مكتبة** طبيب محدد

نص الاستشارة

لبض سريع أحياناً عند صعود الدرج مع تعب عام. هل أحتاج تخطيط قلب؟

توصيات الطبيب — د. محمد الخالدي

سجلي النبض في الراحة والجهود ليومين. وراجعي لإجراء تخطيط قلب. استمري على الأدوية الحالية. أي ألم صدر أو ضيق تنفس شديد → الطوارئ فوراً.

عرض التفاصيل ←

#1 مکتبہ ارسال عام

نص الاستشارة

أعاني من صداع مستمر منذ أسبوعين مع دوخة خفيفة، خاصة عند الوقوف. هل يمكن مراجعة التحاليل المرفقة؟

توصيات الطبيب — د. محمد الخالدي

بعد مراجعة التحاليل والأعراض، يبدو أن الصداع مرتبط بارتفاع ضغط الدم الخفيف. أصبح بمناخة قياس الضغط يومياً، والاستمرار على العلاج الحالي. إذا استمر الصداع أكثر من أسبوعين إضافية، يُفضل زيارة طوارئ أو إعادة التقييم.

عرض التفاصيل ←

Figure 46 Patient consultations list and status tracking



تسجيل الخروج تنبيه تفاصيل الاستشارة لوحة التحكم الاستشارات الاطباء الملف الطبي كلمة المرور

أرسلت 21 يوليو 2026 في 04:44 م مكتلة بانتظار طبيب #1

سؤال الاستشارة

أعاني من صداع مستمر منذ أسبوعين مع دوخة خفيفة، خاصة عند الوقوف. هل يمكن مراجعة التحاليل المرفقة؟

الحالة: مكتملة

المرفقات الطبية

2 ملف(ات) مرفقة مع الاستشارة

تحميل pdf تحاليل دم pdf 125.0 KB

تحميل

شهادة مهنية

وجز أشعة صدر

GameCare Connect — Demo

تحميل

أشعة صدر وجز صورة 500.0 KB

الدكتور/ة: د. محمد الخالدي

المحادثة

إشعار الاستشارة بين المراجع والطبيب

الطبيب — د. محمد الخالدي

بعد مراجعة التحاليل والأعراض، يبدو أن الصداع مرتبط بارتفاع ضغط الدم الخفيف. أوصي بمراقبة قياس الضغط يومياً، والاستمرار على العلاج الحالي.

مراقبة الرد

اكتب سؤالاً متابعة أو توضيحاً للطبيب...

إرسال الرد

[العودة إلى قائمة الاستشارات](#)

Figure 48 Physician consultation review and recommendation form

6.4.6 Medical Reports Module

The Medical Reports Module allows patients to upload supporting medical documents, including medical images, laboratory reports, and PDF files, which are linked directly to consultation records. These documents are stored securely and can be viewed or downloaded only by authorized physicians involved in the consultation, supporting more accurate medical assessment and recommendations.

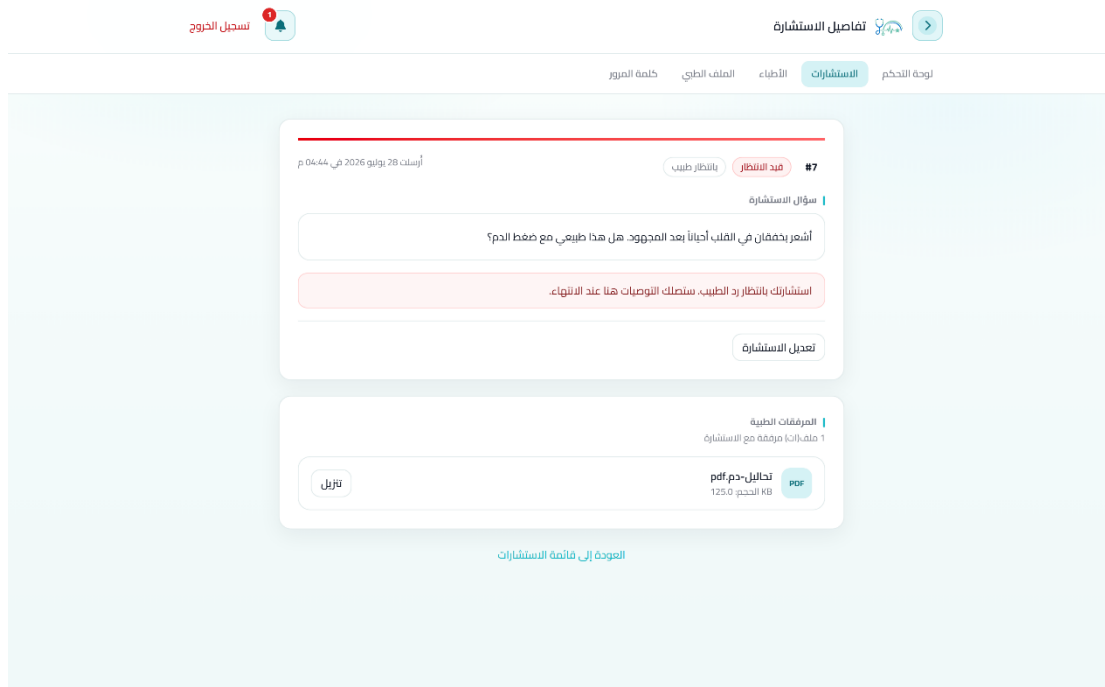


Figure 49 consultation with medical files

6.4.7 Follow-up Messaging Module

The Follow-up Messaging Module provides secure communication between patients and physicians after the physician submits the initial consultation response. Both parties can exchange follow-up messages within the same consultation thread, allowing additional questions and clarifications without creating a new consultation. All messages are stored permanently and associated with the corresponding consultation record.

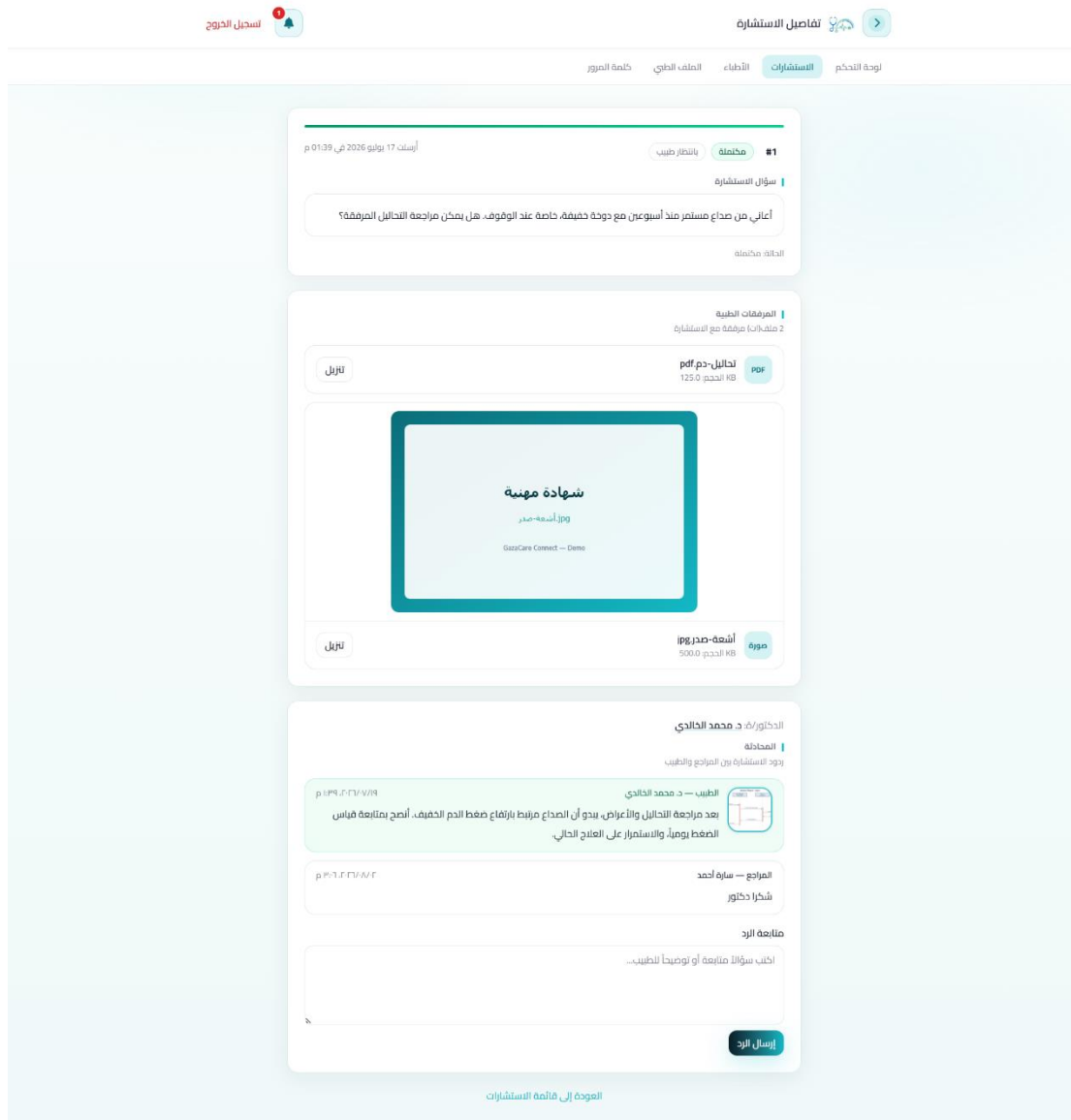


Figure 50 consultation response and messages

6.4.8 Notification Module

The Notification Module delivers real-time in-application for important system events, including physician verification decisions, consultation assignments, physician responses, and follow-up messages. Users can access the notification center to review unread notifications and mark them as read, improving awareness of consultation progress without requiring continuous manual monitoring.

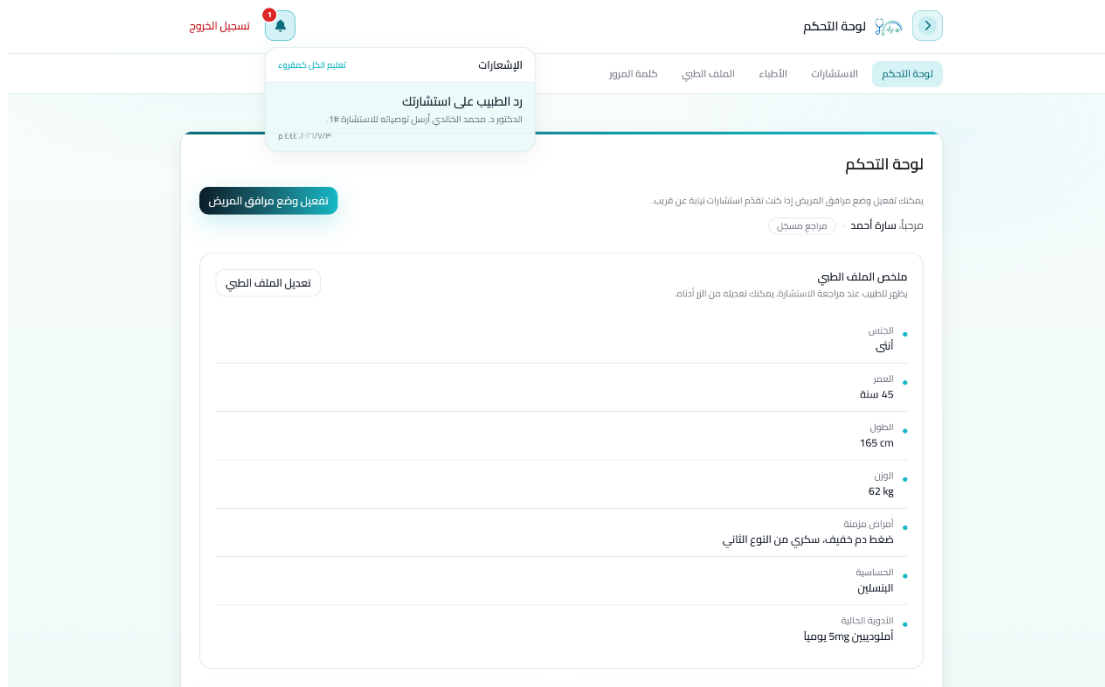


Figure 51 notifications

6.4.9 Administration Module

The Administration Module provides administrators with comprehensive management capabilities for the platform. Administrators can review registered users, monitor account status, and disable or re-enable user accounts when necessary. They are also responsible for reviewing physician registration requests, examining submitted credentials and specialty information, and approving or rejecting physician verification requests. Only verified physicians are permitted to accept and respond to patient consultations.

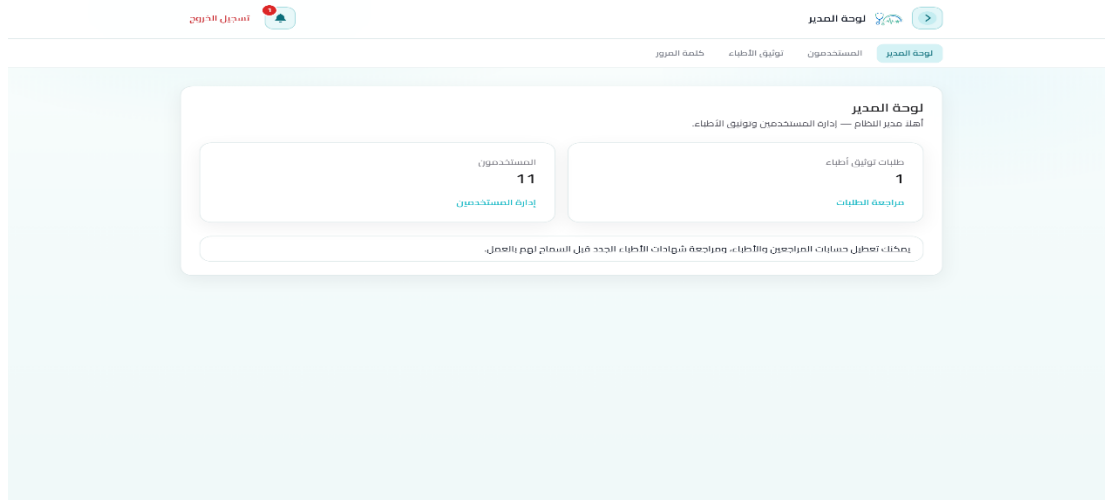


Figure 52 admin dashboard

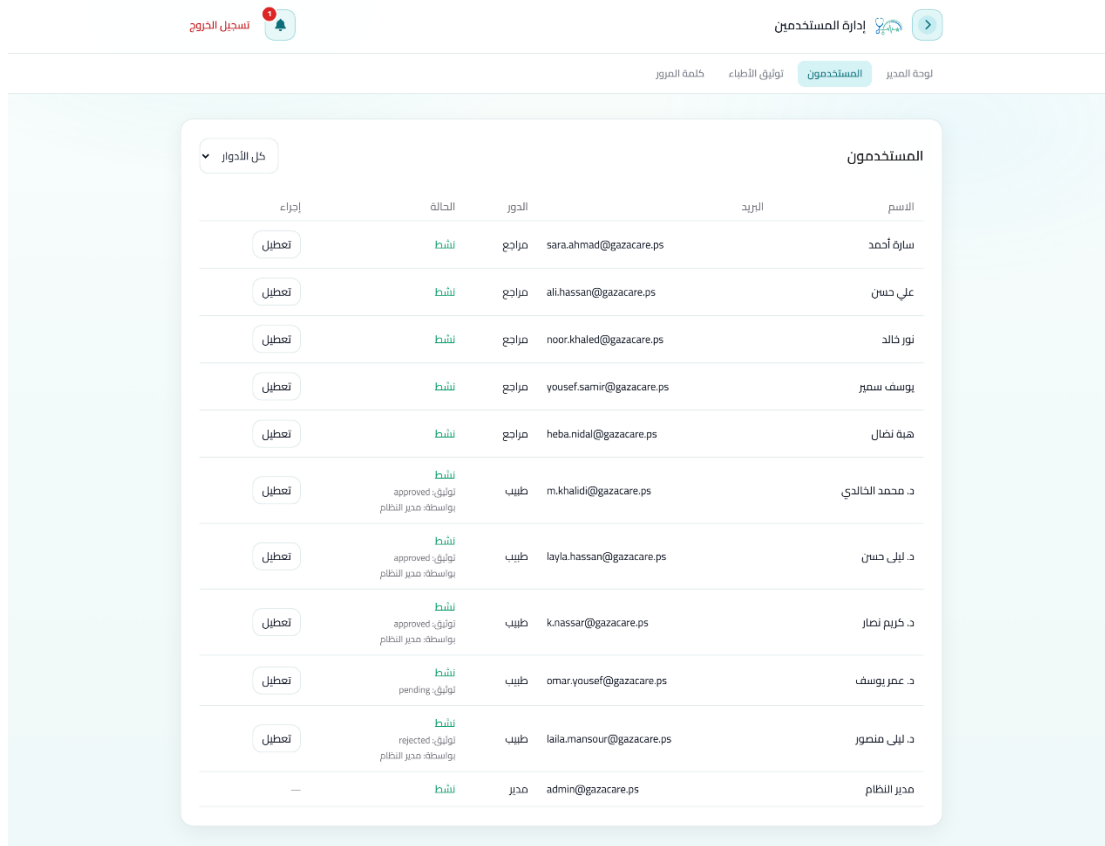


Figure 53 admin manage users

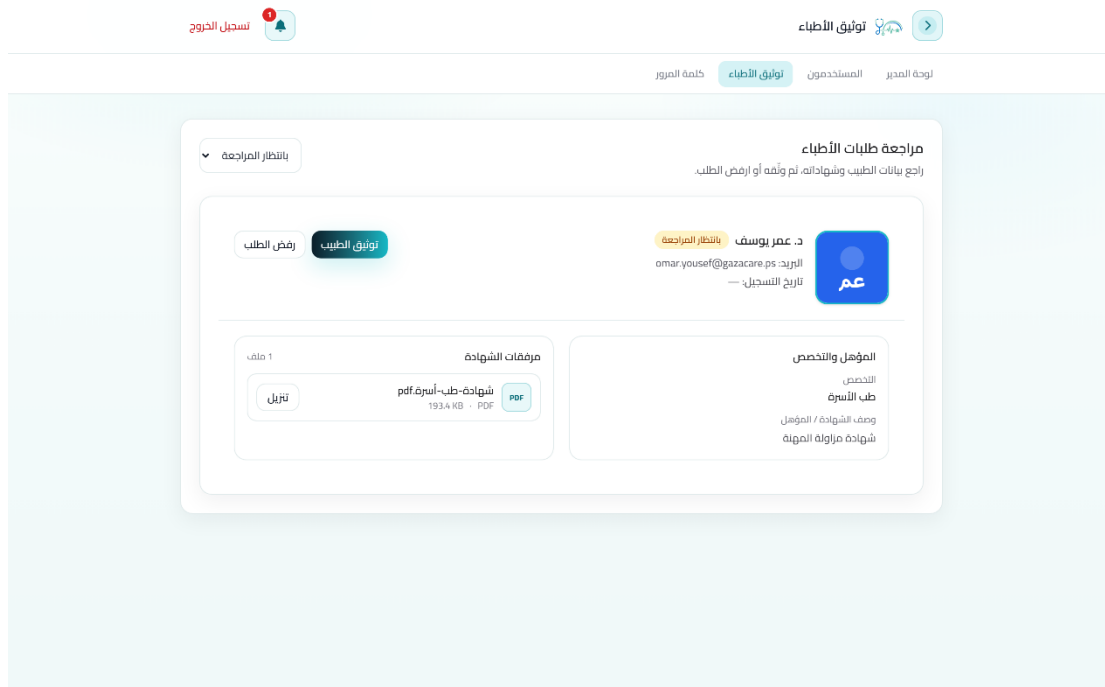


Figure 54 admin pending physicians

6.4.10 Dashboard Module

The Dashboard Module provides customized interfaces for patients, physicians, and administrators based on their assigned roles. Patients have quick access to consultations, physician search, profile management, and notifications. Physicians can manage their professional profiles, consultation requests, and ongoing consultations through their dashboard. Administrators can monitor platform activity, manage users, and oversee physician verification requests from a centralized administrative dashboard.

Role-based access control is enforced throughout all dashboard interfaces, ensuring that each user can access only the functionalities and information associated with their authorized role while maintaining the security and privacy of system data.

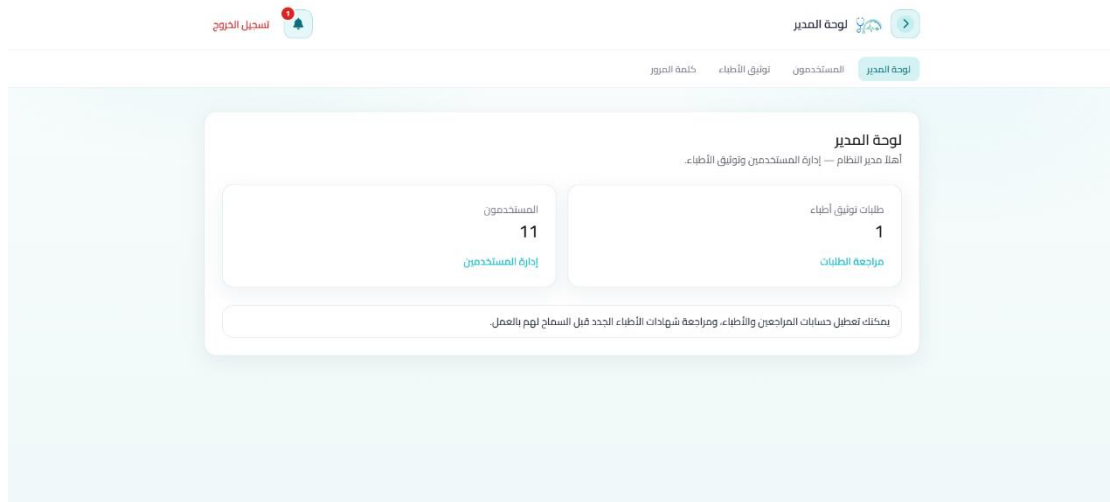


Figure 55 Admin Dashboard



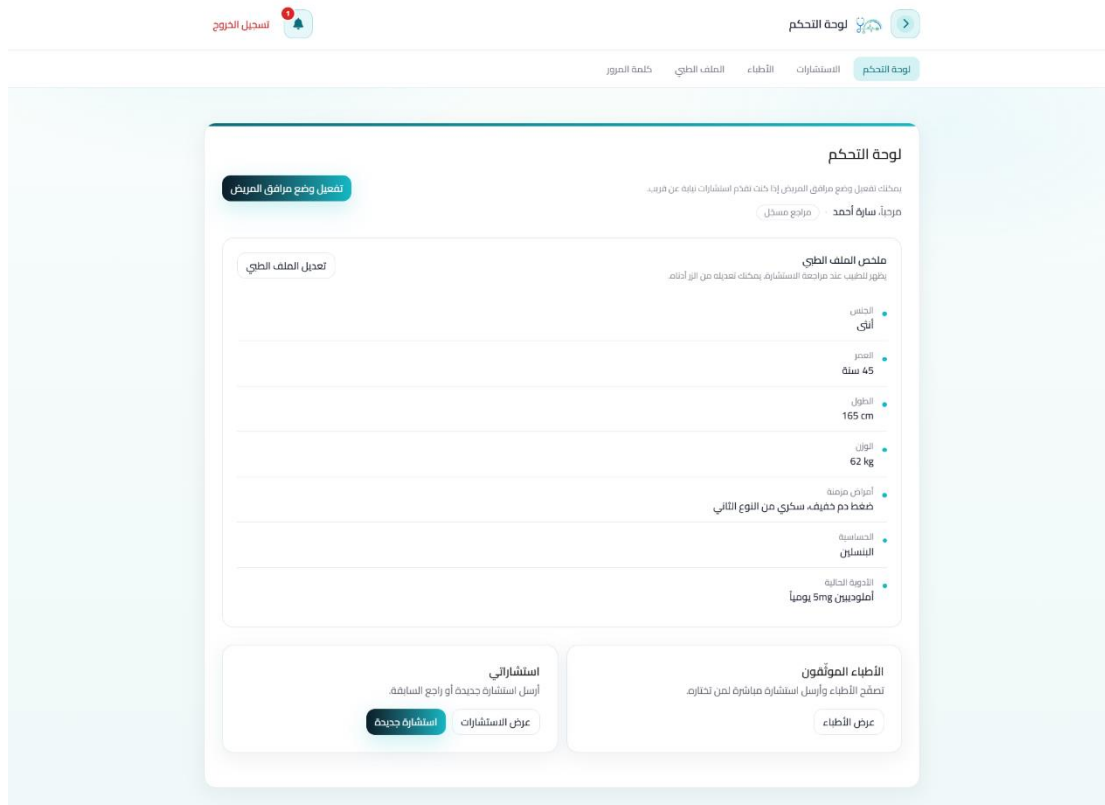


Figure 59 Patient Dashboard

6.5 User Interface

The user interface of GazaCare Connect was designed to provide a simple, intuitive, and responsive experience for all users. Since the platform targets patients, doctors, and administrators with different technical backgrounds, particular attention was given to usability, accessibility, and ease of navigation.

The interface was implemented using React.js and Next.js to create a modern and responsive web application that works efficiently across different screen sizes. A clean layout, consistent color scheme, and well-organized navigation menus were adopted to improve the overall user experience.

For patients, the interface provides easy access to medical profile management, consultation submission, file uploading, and viewing doctor recommendations. The consultation process was designed as a straightforward workflow that minimizes the number of required steps while maintaining clarity.

The doctor interface focuses on efficiency by providing quick access to consultation requests, patient medical information, uploaded reports, and recommendation

submission. The dashboard enables physicians to manage their consultations through a clear and organized workspace.

The administrator interface provides management tools for monitoring users and maintaining the platform. Administrative functions are presented through a simple dashboard that allows efficient system management while preserving security and access control.

Special consideration was given to responsive design, ensuring that the platform can be accessed from desktop computers, tablets, and mobile devices without affecting usability or performance.

The following figures illustrate some of the main user interfaces implemented in the GazaCare Connect system.

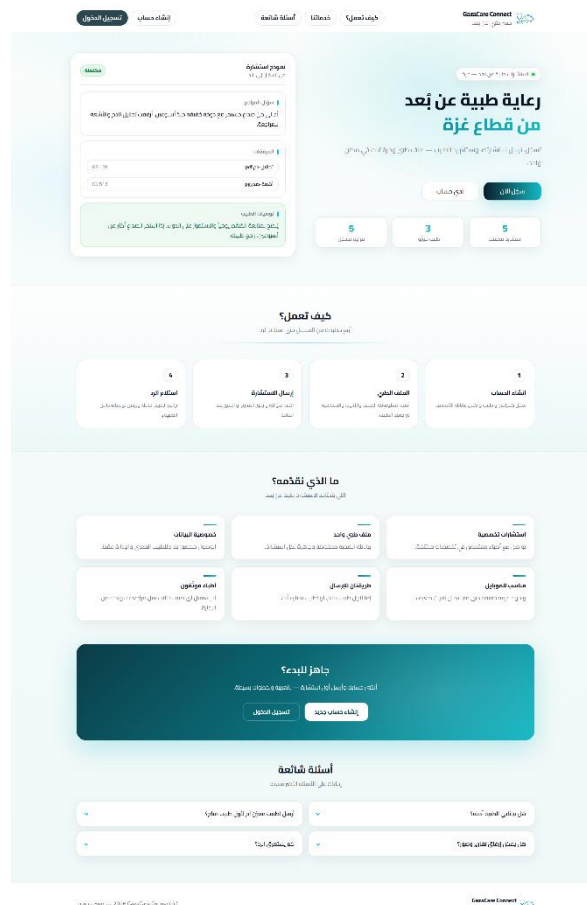


Figure 60 Home Page

6.6 Use Case Implementation

The use cases identified during the analysis and design phases were successfully implemented and integrated into the GazaCare Connect platform. Each use case was developed according to the specified functional requirements, ensuring that patients, physicians, and administrators can perform their tasks efficiently, securely, and according to their assigned roles.

The implemented functionality covers the complete system workflow, beginning with user registration and authentication, followed by consultation submission and physician management, and concluding with administrative supervision. Common functionalities such as login, password management, profile updates, notifications, and logout are available to all authenticated users. In contrast, role-specific functionalities are provided through a role-based access control (RBAC) mechanism, ensuring that each user can access only the features relevant to their role.

The implemented use cases are summarized in Table.

Table 28 Implemented Use Cases

Use Case ID	Use Case	Description	Status
UC-01	Login	Authenticate users and grant role-based access.	Implemented
UC-02	Reset Password	Reset a forgotten password through the registered email address.	Implemented
UC-03	Change Password	Update the current password while authenticated.	Implemented
UC-04	Update Profile	Modify personal and role-specific profile information.	Implemented
UC-05	Receive Notifications	Receive and manage in-application notifications.	Implemented
UC-06	Logout	End the current user session securely.	Implemented

Use Case ID	Use Case	Description	Status
UC-07	Register as Patient	Create a new patient account.	Implemented
UC-08	Enable Caregiver Mode	Enable caregiver mode and specify the relationship type.	Implemented
UC-09	Browse Physicians by Specialty	Browse and filter verified physicians by medical specialty.	Implemented
UC-10	View Physician Details	Display detailed information about a verified physician.	Implemented
UC-11	Submit Consultation	Submit a consultation to either the shared queue or a selected physician.	Implemented
UC-12	Upload Medical Reports	Upload and attach medical reports to a consultation.	Implemented
UC-13	View Consultation Status and Physician Response	View consultation status, physician recommendations, and conversation history.	Implemented
UC-14	Send Follow-up Message (Patient)	Send follow-up messages after receiving the physician's initial response.	Implemented
UC-15	Register as Physician	Create a physician account pending administrator verification.	Implemented
UC-16	View Consultation Requests	View queue, direct, in-progress, and completed consultations.	Implemented
UC-17	Claim Queue Consultation	Claim a pending consultation from the shared consultation queue.	Implemented
UC-18	View Patient Details	Access patient profile and medical information related to a consultation.	Implemented
UC-19	Download / View Medical Reports	View or download medical files attached to a consultation.	Implemented
UC-20	Send Medical Recommendation	Submit a medical recommendation together with the case severity level.	Implemented
UC-21	Send Follow-up Message (Physician)	Continue communication with the patient through the secure consultation thread.	Implemented

Use Case ID	Use Case	Description	Status
UC-22	View Previous Cases	Browse completed and previous consultations.	Implemented
UC-23	Manage Users	View and manage user accounts and account status.	Implemented
UC-24	Disable User Account	Disable or re-enable user accounts when required.	Implemented
UC-25	Approve / Reject Physician	Review physician credentials and approve or reject verification requests.	Implemented

The implemented use cases were fully integrated with both the database and the user interface to provide a complete end-to-end workflow. When a patient submits a consultation, either to the shared consultation queue or directly to a verified physician, the consultation details and any attached medical reports are securely stored in the database. Authorized physicians can then review pending consultations, claim queue cases when appropriate, examine the patient's medical information and uploaded files, provide medical recommendations with an associated case severity level, and continue communication through the secure messaging system. Throughout the consultation process, patients receive notifications regarding consultation updates and can review physician responses and submit follow-up messages within the same consultation thread.

Administrative functionalities are also fully integrated into the platform. Administrators are responsible for managing user accounts, enabling or disabling accounts when necessary, and reviewing physician verification requests before physicians are permitted to provide medical consultations.

Role-based authorization is enforced throughout the system to ensure secure access to platform resources. Patients are limited to managing their own profiles and consultations, verified physicians can access only consultations assigned or claimed by them, and administrators are granted management privileges without participating in the medical consultation process. This implementation ensures data security, maintains user privacy, and enforces appropriate separation of responsibilities across the platform.

6.7 Chapter Summary

This chapter presented the implementation of the GazaCare Connect platform, translating the design and modeling work of the previous chapter into a functional telemedicine system. The development environment, based on React.js, Next.js, and TypeScript for the front end and Laravel with PHP for the back end, was described together with the three-tier architecture that separates the presentation, application, and data layers.

The MySQL database implementation was detailed, including the consolidated users table with role-based differentiation and the linked profile, consultation, messaging, file, and notification tables. The chapter also described the ten core system modules—authentication, user profile, caregiver mode, physician directory, consultation, medical reports, follow-up messaging, notifications, administration, and dashboard—along with the user interface design for patients, physicians, and administrators, and the implementation of the previously defined use cases.

Together, these implementation details demonstrate how the analysis, design, and modeling decisions made in earlier chapters were translated into a secure, modular, and role-based platform capable of supporting asynchronous medical consultations for patients in Gaza.

CHAPTER 7 TESTING

7.1 Introduction

A multi-level testing strategy was applied throughout the development of GazaCare Connect to verify functional correctness, system integration, usability, and security. Testing was conducted incrementally in alignment with the Incremental Development Model adopted in Chapter 3. Each development increment was tested before integration with subsequent modules.

7.2 Unit Testing

Unit tests were written to verify the correctness of individual backend functions and API endpoints in isolation. Laravel's built-in PHPUnit testing framework was used for this purpose. The following units were covered:

User Registration Validation:

Tests verified that the registration endpoint correctly rejects missing required fields (name, email, password, role), duplicate emails, and invalid email formats. Tests also confirmed that valid registration data results in a new user record and a returned authentication token.

Authentication Token Behavior:

Tests confirmed that protected routes return HTTP 401 (Unauthorized) when accessed without a token, and that a valid token grants the expected access. Logout tests verified that the token is invalidated server-side after logout.

Medical Profile Operations:

Tests verified that a patient could create and update a medical profile, and that the profile data (height, weight, chronic conditions, allergies, medications, and history) was correctly stored and retrieved. Tests also confirmed that a doctor cannot access or modify a patient's medical profile directly.

Consultation Submission:

Tests verified that the consultation creation endpoint correctly rejects descriptions shorter than 10 characters, accepts valid input, sets the initial status to Pending, and correctly links uploaded file attachments to the consultation record.

Doctor Response Submission:

Tests verified that only the assigned doctor can submit a response to a consultation, that the response text is saved correctly, and that the consultation status transitions to Completed upon successful response submission.

7.3 Integration Testing

Integration testing verified that the different modules of the system work correctly together and that the frontend communicates with the backend API as expected. The following integration scenarios were tested:

End-to-End Consultation Flow:

The complete consultation lifecycle was traced from patient registration, profile creation, consultation submission, doctor acceptance, response submission, and final consultation retrieval by the patient. Each step was verified to produce the correct database state and frontend display.

File Upload and Retrieval Integration:

Tests confirmed that files uploaded during consultation submission are stored on the server, correctly associated with the patient and consultation records in the database, and retrievable by the doctor through the protected download endpoint. Unauthorized file access attempts were confirmed to return HTTP 403 (Forbidden).

Role-Based Access Control Integration:

Cross-role access attempts were tested systematically. Patient accounts attempting to access the doctor's waiting queue, or doctors attempting to modify another doctor's active consultation, were confirmed to receive appropriate HTTP 403 rejection responses.

Frontend–Backend Error Handling:

Integration tests verified that when the backend returns error responses (such as validation failures or server connection issues), the frontend properly handles these responses and displays user-friendly Arabic error messages instead of raw technical error codes.

For example, when the server is unreachable, the registration form displays the message: "فشل الاتصال بالخادم. تحقق من الإنترنت."

This ensures that system errors are communicated clearly to end users without exposing technical details.

7.4 Usability Testing

Usability testing was conducted to evaluate whether end users could navigate and operate the platform effectively without technical guidance. The testing focused on the two primary user groups: patients and doctors.

Task Completion Testing:

Representative users completed key tasks independently, including account creation, medical profile completion, consultation submission with file attachment, and viewing completed consultation responses. All tasks were completed successfully without technical assistance.

Arabic Language and RTL Layout Verification:

The interface was verified to render correctly in right-to-left (RTL) orientation across all tested screens. Form fields, labels, buttons, menus, and error messages displayed correctly in Arabic.

Mobile Responsiveness:

All interface screens were tested on desktop and mobile viewport sizes. The layout adapted correctly to smaller screens without horizontal scrolling or overlapping elements.

Error Message Clarity:

Feedback messages for common errors—including empty required fields, invalid email addresses, and server connection failures—were reviewed to ensure they were presented in clear, understandable Arabic suitable for non-technical users.

7.5 Security Testing

Given the sensitivity of medical data, security testing was a priority throughout the development lifecycle. The following security checks were performed:

Authentication Enforcement:

All API routes handling patient data, consultations, and file access were protected using Laravel Sanctum authentication middleware. Requests without valid tokens correctly returned HTTP 401 (Unauthorized).

Password Storage:

Database inspection confirmed that all passwords are stored exclusively as bcrypt hashes. No plain-text passwords exist in the database.

Authorization Boundary Testing:

Tests confirmed that patients cannot access or modify another patient's medical profile or consultations. Doctors cannot respond to consultations assigned to other physicians, and unauthenticated users cannot access protected resources.

File Access Protection:

Uploaded medical files are stored outside the public directory. File retrieval is performed through protected API endpoints that verify user authentication and

ownership before granting access. Unauthorized access attempts correctly returned HTTP 403 (Forbidden).

Input Validation:

All API endpoints apply Laravel server-side validation rules. Invalid inputs—including empty fields, incorrect data types, and excessively long strings—were correctly rejected with descriptive validation messages.

HTTPS Communication:

The system is configured to operate over HTTPS/SSL in the production environment, ensuring that authentication tokens, consultation data, and uploaded files are encrypted during transmission.

7.6 CHAPTER SUMMARY

This chapter presented the multi-level testing strategy applied to verify the correctness, reliability, and security of GazaCare Connect. Unit testing confirmed the correct behavior of individual backend functions and API endpoints, including registration validation, authentication token handling, medical profile operations, and consultation and response submission. Integration testing verified that the frontend and backend modules interact correctly across the complete consultation lifecycle, file upload and retrieval, role-based access control, and error handling.

Usability testing confirmed that patients and physicians could complete key tasks independently, that the Arabic right-to-left interface rendered correctly across screens, and that the platform adapted properly to mobile and desktop viewports. Security testing verified authentication enforcement, bcrypt password hashing, authorization boundaries between roles, protected file access, input validation, and HTTPS-based encrypted communication.

Taken together, the results of unit, integration, usability, and security testing confirm that GazaCare Connect meets its functional and non-functional requirements and operates as a reliable, secure, and accessible platform for asynchronous medical consultations.

CHAPTER 8 CONCLUSION AND RECOMMENDATIONS

8.1 Introduction

This chapter presents the overall conclusion of the GazaCare Connect research and development project. It summarizes how the research problem, objectives, and questions identified in Chapter One were addressed through the literature review, methodology, system design, implementation, and testing presented in the previous chapters. The chapter also highlights the contributions of the study and concludes with recommendations for stakeholders and future development directions.

8.2 Conclusion

The GazaCare Connect project was developed in response to an identified need in the healthcare context of the Gaza Strip: a secure, organized, and integrated electronic platform that enables patients to access remote specialized medical consultations in a reliable and confidential manner. This need was established through an analysis of the local healthcare context, a review of existing telemedicine platforms, and a structured survey of 65 participants that highlighted the reliance on informal communication channels for medical advice.

Guided by these findings, the system was developed using an Incremental Development Model, allowing the platform to evolve through repeated cycles of analysis, design, implementation, and testing. Each functional increment, from user authentication to consultation management, was verified before integration with the rest of the system. The platform was modeled using standard UML diagrams, including a use case diagram, sequence diagrams, and a class diagram, which provided a clear blueprint prior to implementation.

The implementation stage translated this design into a working three-tier web application using React.js, Next.js, and TypeScript for the frontend, Laravel and PHP for the backend, and MySQL as the relational database management system.

The platform was organized into integrated functional modules covering authentication, patient and physician profiles, caregiver mode, physician directory, consultation management, medical file handling, follow-up messaging, notifications, administration, and role-based dashboards. Together, these modules support the

complete consultation lifecycle, from patient submission to physician review and recommendation.

A comprehensive testing strategy, including unit testing, integration testing, usability testing, security testing, confirmed that the system satisfies its functional requirements, operates reliably across desktop and mobile devices, correctly enforces role-based access control, and protects sensitive medical data through secure authentication, hashed password storage, and controlled access to uploaded medical files.

Overall, GazaCare Connect provides patients in the Gaza Strip with a secure and structured means of accessing specialized physicians remotely while reducing dependence on informal communication channels that may expose patients to misinformation, inconsistent medical advice, and privacy risks.

8.3 Achievement of the Research Objectives

The research objectives defined in Chapter One were successfully achieved through the design, implementation, and evaluation of the GazaCare Connect platform. Specifically:

- A secure web-based platform for remote specialized medical consultations was successfully designed and implemented.
- Secure authentication, hashed password storage, and role-based access control were implemented to protect user accounts and medical information.
- Patients can access verified physicians regardless of geographical limitations within the Gaza Strip.
- The asynchronous consultation workflow supports both queue-based and direct physician assignment, improving access to specialized medical advice.
- Comprehensive medical profiles and uploaded medical reports enable physicians to make more informed recommendations.
- Dedicated consultation management and follow-up messaging support continuous communication between patients and physicians.
- Consultation history and persistent medical profiles contribute to continuity of care.
- All essential functional requirements identified during the analysis phase—including registration, authentication, consultation submission, physician

recommendations, file uploads, notifications, and physician verification—were fully implemented.

- System testing verified the platform's functionality, usability, performance, and security.
- The developed platform provides a structured and secure alternative to informal methods currently used for obtaining medical advice in Gaza.

8.4 Contributions of the Study

This study makes several important contributions from technical, scientific, and practical perspectives.

From a humanitarian perspective, the platform provides patients in the Gaza Strip with a secure and organized method for obtaining specialized medical consultations remotely. This reduces dependence on informal communication channels and improves access to qualified healthcare professionals.

Scientifically, the study presents a practical implementation of an asynchronous telemedicine platform specifically designed for a resource-constrained environment. The project integrates findings from the literature review, system analysis, software engineering principles, and user requirements into a complete development lifecycle.

Technically, the project demonstrates the integration of modern web technologies through a three-tier architecture combining React.js, Next.js, Laravel, PHP, MySQL, Laravel Sanctum, and role-based access control mechanisms. The modular architecture also facilitates future system maintenance and expansion.

From a societal perspective, the platform contributes to improving access to healthcare services while potentially reducing unnecessary hospital visits and supporting more efficient utilization of available medical specialists.

8.5 Scope of the Study

This study focused on the design, implementation, and evaluation of a web-based asynchronous telemedicine platform for remote specialized consultations.

The implemented system includes patient, physician, and administrator roles, secure authentication and account management, medical profiles, caregiver mode, physician

directory, queue-based and direct consultation workflows, medical file attachments, follow-up messaging, physician verification, notifications, and role-based dashboards.

Advanced features such as large-scale deployment, integration with hospital information systems, electronic health records, artificial intelligence support, video consultations, and advanced analytics were intentionally excluded from the current project scope and are recommended for future work.

8.6 Recommendations

Based on the findings of this study, the following recommendations are proposed:

- Healthcare institutions and non-governmental organizations operating in the Gaza Strip should consider adopting structured telemedicine platforms to improve access to specialized healthcare services.
- Medical institutions should establish formal physician verification procedures similar to those implemented in GazaCare Connect to enhance patient trust in remote consultations.
- Policymakers should encourage the adoption of secure digital healthcare solutions as part of broader healthcare modernization initiatives.
- Future developers should continue improving platform accessibility, usability, and operational monitoring as the system grows.

8.7 Recommendations for Future Development

Several enhancements can further improve GazaCare Connect in future versions:

Physician Rating and Feedback System

Implement a structured physician rating and feedback mechanism that allows patients to evaluate consultation quality after each completed consultation.

Advanced Physician Search

Expand physician search capabilities by allowing filtering according to specialty, availability, consultation experience, and future patient ratings.

Administrative Analytics Dashboard

Develop a comprehensive analytics dashboard presenting key performance indicators such as consultation volume, physician response time, physician activity, and consultation completion rates.

Mobile Application

Develop a dedicated mobile application using React Native or Progressive Web Application (PWA) technologies to improve accessibility for smartphone users.

Multi-language Support

Support both Arabic and English interfaces to facilitate collaboration with physicians located outside Palestine while maintaining Arabic as the primary language.

Dark Mode

Implement a complete dark mode interface to improve user comfort during nighttime use.

Video Consultation Support

Extend the platform by integrating secure video consultations for cases requiring direct interaction between physicians and patients.

Artificial Intelligence Assistance

Incorporate AI-based tools to assist physicians by prioritizing consultations according to urgency, suggesting possible diagnoses, and supporting decision-making while keeping physicians responsible for final medical recommendations.

8.8 Final Remarks

GazaCare Connect demonstrates that a carefully designed and systematically implemented telemedicine platform can provide an effective solution for improving access to specialized healthcare services in the Gaza Strip. By applying modern software engineering principles, secure system architecture, and user-centered design, the project successfully delivers a reliable web-based platform that supports asynchronous medical consultations while ensuring privacy, security, and ease of use.

The implementation and evaluation results confirm that the platform satisfies the identified functional and non-functional requirements and provides a solid foundation for future expansion. With continued development and institutional adoption, GazaCare Connect has the potential to become an important component of digital healthcare services in Palestine and contribute to improving healthcare accessibility for patients facing geographical, logistical, and resource-related challenges.

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Appendices

This section provides supplementary materials that support and extend the information presented throughout this report, including project statistics, platform comparisons, source code repositories, a live demo deployment, the system test plan, and the UI/UX design files.

Appendix A: Project Statistics (Chapter 3)

The full dataset and statistical analysis referenced in Chapter 3 (Research Methodology) are available at the following link:

https://docs.google.com/spreadsheets/d/1RtWErAtQlgQwOB8UUtPLV7tVUgr8jahN/edit?usp=drive_link&ouid=116200435113213369578&rtpof=true&sd=true

Appendix B: Related Platforms (Chapter 2) with Documented Limitations

Documentation of the links and platforms referenced in Chapter 2 (Literature Review), including a record of their observed limitations, is available at the following link:

<https://drive.google.com/file/d/1XqZz6WIOdI0pDTbi9WIBoUOdjeXV1JYV/view?usp=sharing>

Appendix C: Source Code Repositories

The GazaCare Connect source code is available on GitHub:

Backend repository: <https://github.com/lamadirbi/GazaCareConnect-backend>

Frontend repository: <https://github.com/lamadirbi/GazaCareConnect-frontend>

Appendix D: Live Frontend Demo

A live simulation of the frontend, populated with sample data and deployed on Netlify, can be accessed at:

<https://gazacareconnect.netlify.app/>

Appendix E: System Test Plan

The full test plan for the GazaCare Connect system is available at the following link:

<https://drive.google.com/file/d/19TrhEm1MVjwikXOv5UI4GDDuasFbqYOn/view?usp=sharing>

Appendix F: UI/UX Design (Figma)

The UI/UX design files for GazaCare Connect can be viewed on Figma at:

<https://www.figma.com/design/R9aQTe6Q0VIsQ0TCX7GdZe/gazacareconnect?node-id=5-584&t=QEazAm5Bt7hmu4Wx-1>